MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title: CALCULATION
49WF BORE WELL PUMP HOUSE 3 & 4
DESIGN CALCULATION

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor.
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities

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1. INTENT

This document is to prepare design of Borewell pump station 3 & 4. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the structure.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis and design of beam and column members have been carried out in STAAD PRO.

For foundation design, force values are directly taken from STAAD output & designed in excel sheet.

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4. DESIGN REQUIREMENTS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

4.3 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Foundation	75
Exposed to weather or contact with ground	Column	50

4.4 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundation design has been considered 350 Kn/m^2 at 1.5 m below FGL

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5. DESIGN LOADS

5.1 Dead Load

5.1.1 Roof Dead Loads

- Provision for Electrical Lighting = 0.05 kN/m²

Ceiling

Suspended steel channel system = 0.10 kN/m²

Acoustical Fiber Tile (16mm) = 0.05 kN/m²

Sub-total = 0.2 kN/m²

- Hollow Core Slab (PCI 6th Edition)

(HCS-200 Topped)(To be confirmed by Vendor) = 3.1 kN/m²

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Prestressed Hollowcore Floor

OVERALL STRUCTURAL DEPTH MM	SELF WT. KN/M²	NO. OF CORES IN UNIT	SPANS INDICATED BELOW ALLOW FOR CHARACTERISTIC SERVICE LOAD (LIVE LOAD) PLUS SELF WEIGHT PLUS 1.5KN/M² FOR FINISHES CHARACTERISTIC SERVICE LOADS KN/M²							
			0.75	1.50	2.00	2.50	3.00	4.00	5.00	10.0
			Clear span in metres							
100	2.38	0	5.39	5.19	5.08	4.91	4.72	4.40	4.14	3.30
150	2.52	11	7.40	7.40	7.40	7.40	7.28	6.80	6.40	5.12
150 (Dense Slab)	2.94	11	7.40	7.40	7.40	7.35	7.08	6.64	6.27	5.06
200	3.10	11	9.40	9.40	9.40	9.40	9.26	8.69	8.21	6.65
250	3.43	6	10.80	10.80	10.80	10.80	10.80	10.80	10.56	8.60
300	3.88	6	13.50	13.50	13.50	13.50	13.18	12.44	11.81	9.70
350	4.23	6	15.00	15.00	15.00	15.00	14.89	14.08	13.39	11.06
400	4.57	6	16.00	16.00	16.00	16.00	15.97	15.13	14.42	11.98
450	4.92	6	18.00	18.00	18.00	17.70	17.19	16.33	15.58	13.02
500	5.27	6	18.00	18.00	18.00	18.00	18.00	17.13	16.38	13.76

Built-in Water Proofing System

100mm thk Polystyrene Foam

0.01 kN/m² per 25.4mm thickness x (100/25.4mm) = 0.039 kN/m²

Water proofing membrane

Bituminous, smooth surface = 0.07 kN/m²

90 mm thick average cement foam concrete screed Lightweight

Concrete 0.38 kN/m² per 25mm thickness x (90/25mm) = 1.368 kN/m²

75 mm Concrete screed above HCS = 21 x 0.075 =1.6 kN/m²

Sub-total = 3.1 kN/m²

Total Load = 6.4 kN/m²

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UDL on Edge Ledger Beams (RB1) = 6.4 x 4.2/2 = 13.45 Kn/m

Moment on all the External Beams due to Cantilever Portion of slab and Screed

= (0.15x24+(0.05+0.2)/2x20)*0.6*0.6/2 = 1 knm/m

5.1.2 Ground Floor Wall Loads

- 250 thk. Outer CMU (Partially Grouted Cell @ 800 c-c)

  16 x 0.25 x (3.5-0.5)m clear height = 12 kN/m

5.2 Live Load

5.2.1 Roof Live Loads

Live loads were taken from Design Criteria clause no 4.2.2.

Roof not accessed for maintenance = 0.5 kN/m²

Udl Due to Live Load on the Edge Ledger Beam (RB1) = 0.5 x 4.2/2 = 1.05 kN/m

Moment on all the External Beams due to Live Load on

Cantilever Portion of slab = 0.5 x 0.6 x 0.6/2 = 0.1 kN/m /m

5.2.2 Monorail Load

Load due to 1Ton capacity monorail with manual hoist = 10 kN

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6 WIND LOAD CALCULATION

Velocity Pressure , qz evaluated at height z shall be calculated by the following equation:

$qz = 0.0473 \times 10^{-3} K_z K_{zt} K_d V^2 I$ kN/m^2 (Refer Cl. No 7.2.8 of SBC-301)

Basic Wind Speed	V =	177	km/h
Velocity pressure exposure coefficient	Kz =	1.04	As per Table 7.2-2(Page-7/44)- z = 0 to 5m of SBC-301
Exposure Categories.	=	D	As per Project Basis of Design
Topographic Factor	K_{zt}=	1.00	As per Cl. 6.4.3.2

(As per Pt no 4 of cl 6.4.3.1 of SBC-301 , $H/L_h < 0.2$, hence topographic effect not considered)

Wind Directionality Factor	K_d =	0.85	As per Table 6.4-1(Page-6/7)	
Importance factor	I =	1.00	As per Table 6.5-1(Page-6/9)	
Hence, for z = 10m	qz =	1.31	Kn/m2 =	0.131 Ton/m2

Design wind pressure on structure is ; p = qh (GCpf - GCpi) (Refer Eq. 7.2-15 of SBC-301)

GCpf = External pressure co-efficient (Refer Figure No. 7.2 - 6 of SBC-301)
GCpi = Internal pressure co-efficient (Refer Figure No. 7.2 - 1 of SBC-301)

GCpi = ±0.18

WIND ALONG X DIRECTION

GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)
Higher values are adopted for conservative design

Wind load for GCpi = + 0.18 =	1.31 x (0.61-0.18) =	0.56	Kn/m2	(Windward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.43-0.18) =	-0.80	Kn/m2	(Leeward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.45-0.18) =	-0.83	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 1.6m =	1.6 x 0.56 =	0.90	Kn/m
Load on columns for span 4.9m =	4.9 x 0.56 =	2.74	Kn/m
Load on columns for span 3.3m =	3.3 x 0.56 =	1.85	Kn/m

Load on Leeward columns :

Load on columns for span 1.6m =	1.6 x (-0.80) =	-1.28	Kn/m
Load on columns for span 4.9m =	4.9 x (-0.80) =	-3.92	Kn/m
Load on columns for span 3.3m =	3.3 x (-0.80) =	-2.64	Kn/m

Load on side wall columns :

Load on columns for span 2.1m =	2.1 x (-0.83) =	-1.74	Kn/m

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GCpf = 0.61 (Windward direction), -0.43 (Leeward direction), -0.45 (Side walls) — Higher values are adopted for conservative design

Wind load for GCpi = - 0.18 =	1.31 x (0.61+0.18) =	1.03	Kn/m2	(Windward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.43+0.18) =	-0.33	Kn/m2	(Leeward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.45+0.18) =	-0.35	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 1.6m =	1.6 x 1.03 =	1.65	Kn/m
Load on columns for span 4.9m =	4.9 x 1.03 =	5.05	Kn/m
Load on columns for span 3.3m =	3.3 x 1.03 =	3.40	Kn/m

Load on Leeward columns :

Load on columns for span 1.6m =	1.6 x (-0.33) =	-0.53	Kn/m
Load on columns for span 4.9m =	4.9 x (-0.33) =	-1.62	Kn/m
Load on columns for span 3.3m =	3.3 x (-0.33) =	-1.09	Kn/m

Load on side wall columns :

Load on columns for span 2.1m =	2.1 x (-0.35) =	-0.74	Kn/m

WIND ALONG Z DIRECTION

GCpf = 0.61 (Windward direction), -0.43 (Leeward direction), -0.45 (Side walls) — Higher values are adopted for conservative design

Wind load for GCpi = + 0.18 =	1.31 x (0.61-0.18) =	0.56	Kn/m2	(Windward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.43-0.18) =	-0.80	Kn/m2	(Leeward direction)
Wind load for GCpi = + 0.18 =	1.31 x (-0.45-0.18) =	-0.83	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 2.1m =	2.1 x 0.56 =	1.18	Kn/m

Load on Leeward columns :

Load on columns for span 2.1m =	2.1 x (-0.80) =	-1.68	Kn/m

Load on side wall columns :

Load on columns for span 1.6m =	1.6 x (-0.83) =	-1.33	Kn/m
Load on columns for span 4.9m =	4.9 x (-0.83) =	-4.07	Kn/m
Load on columns for span 3.3m =	3.3 x (-0.83) =	-2.74	Kn/m

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GCpf = 0.61 (Windward direction)
-0.43 (Leeward direction)
-0.45 (Side walls)

Higher values are adopted for conservative design

Wind load for GCpi = - 0.18 =	1.31 x (0.61+0.18) =	1.03	Kn/m2	(Windward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.43+0.18) =	-0.33	Kn/m2	(Leeward direction)
Wind load for GCpi = - 0.18 =	1.31 x (-0.45+0.18) =	-0.35	Kn/m2	(Side walls)

Load on windward columns :

Load on columns for span 2.1m = 2.1 x 1.03 = 2.16 Kn/m

Load on Leeward columns :

Load on columns for span 2.1m = 2.1 x (-0.33) = -0.69 Kn/m

Load on side wall columns :

Load on columns for span 1.6m =	1.6 x (-0.35) =	-0.56 Kn/m
Load on columns for span 4.9m =	4.9 x (-0.35) =	-1.72 Kn/m
Load on columns for span 3.3m =	3.3 x (-0.35) =	-1.16 Kn/m

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7. DESIGN OF FOUNDATION MKD. F1 (STAAD Node -81)

	Along-X	Along-Z
Size of column :	0.40m	0.40m
Size of fdn. raft :	1.50m	1.50m
Thickness of raft :	0.40 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
Unit weight of conc :	24 kN/m^3	
Unit weight of soil :	18 kN/m^3	

SYMBOLS AND NOTATIONS

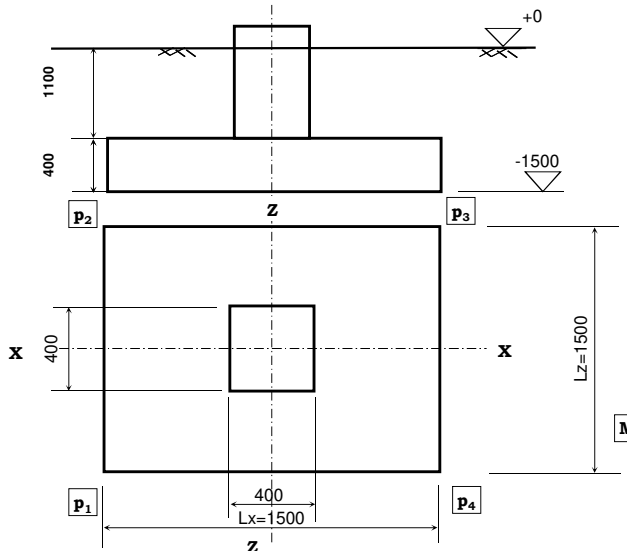
Dead Load	DL
Live Load	LL
Roof Live Load	LR
Wind Load along X Direction Cpi = +0.18	WLX1
Wind Load along X Direction Cpi = - 0.18	WLX2
Wind Load along Z Direction Cpi = +0.18	WLZ1
Wind Load along Z Direction Cpi = - 0.18	WLZ2
Temperature Load	TL

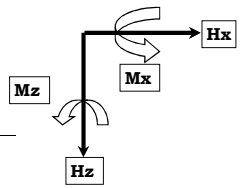
Calculation of Weight of Foundation :

Weight of raft = (1.5*1.5*0.4*24) = 21.60KN
Weight of soil = ((1.5*1.5)-(0.4*0.4))*1.1*18 = 41.38KN
Total = 62.98KN

Design Input:

Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f$_y$=	420 MPa
Clear cover for main reinforcement bars =	50mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	10mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	10mm
Mininum percentage of reinforcement at bottom=	0.18%





Effective depth of slab along X direction = 342mm = 0.342m

Effective depth of slab along Z direction = 326mm = 0.326m

Clear projection along Z direction = 0.55m

Clear projection along X direction = 0.55m

H_X=Force along X-X direction

Hz=Force along Z-Z direction

M_X=Moment about X-X direction

Mz=Moment about Z-Z direction

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Combined Load Cases:		Load at bottom of foundation				
Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)
Allowable Stress Design						
201	1.0DL+1.0LL+1.0TL	232.329	-14.008	5.817	10.914	3.565
202	1.0DL+1.0LR	236.864	-11.713	4.987	14.366	7.972
203	1.0DL+0.75LL+0.75LR+0.75TL	236.333	-13.456	5.617	11.603	4.599
204	1.0DL+1.0WLX1	239.804	-21.465	17.441	14.962	8.351
205	1.0DL+1.0WLX2	239.697	-22.818	17.882	14.925	8.325
206	1.0DL+1.0WLZ1	233.946	-9.304	4.222	9.82	3.415
207	1.0DL+1.0WLZ2	233.822	-10.67	4.655	9.801	3.407
208	1.0DL+0.75LL+0.75LR+0.75WLX1	246.262	-18.94	14.283	13.998	7.92
209	1.0DL+0.75LL+0.75LR+0.75WLX2	246.182	-19.955	14.613	13.97	7.901
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	241.869	-9.819	4.369	10.141	4.219
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	241.776	-10.843	4.693	10.127	4.213
212	0.6DL+1.0WLX1	147.354	-16.798	15.452	9.077	5.108
213	0.6DL+1.0WLX2	147.248	-18.15	15.893	9.04	5.082
214	0.6DL+1.0WLZ1	141.497	-4.636	2.233	3.935	0.172
215	0.6DL+1.0WLZ2	141.372	-6.002	2.666	3.915	0.164
Ultimate Conditions						
101	1.4DL	323.573	-16.337	6.96	20.598	11.35
102	1.4DL+1.7LL+0.5LR	336.242	-16.112	6.848	18.97	10.674
103	1.2DL+1.6LR+0.8WLX1	293.477	-21.91	15.967	17.299	9.708
104	1.2DL+1.6LR+0.8WLX2	293.392	-22.992	16.319	17.269	9.687
105	1.2DL+1.6LR+0.8WLZ1	288.791	-12.181	5.392	13.185	5.759
106	1.2DL+1.6LR+0.8WLZ2	288.692	-13.273	5.738	13.17	5.753
107	1.2DL+1.6LR+1.0LL	292.297	-13.928	5.921	16.244	9.155

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108 1.2DL+1.0LL+1.6WLX1+0.5LR	299.87	-29.553	25.855	17.025	9.692
109 1.2DL+1.0LL+1.6WLX1+0.5LR	299.701	-31.718	26.56	16.965	9.652
110 1.2DL+1.0LL+1.6WLX1+0.5LR	290.499	-10.095	4.704	8.798	1.796
111 1.2DL+1.0LL+1.6WLX1+0.5LR	290.3	-12.281	5.397	8.767	1.783
112 1.2DL+1.0LL	283.112	-13.858	5.895	16.8	9.371
113 0.9DL+1.6WLX1	221.899	-26.176	24.426	13.64	7.686
114 0.9DL+1.6WLX2	221.729	-28.341	25.13	13.581	7.645
115 0.9DL+1.6WLZ1	212.527	-6.718	3.275	5.413	-0.21
116 0.9DL+1.6WLZ2	212.328	-8.903	3.967	5.382	-0.224
117 1.4DL+ 1.4TL+1.7LL+0.5LR	329.861	-19.59	8.13	14.849	4.816

Check for overturning and sliding:

Load Case	Load Combination	V (kN)	Hx (kN)	Mz (kN-m)	Hz (kN)	Mx (kN-m)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
201	1.0DL+1.0LL+1.0TL	232.3	-14.0	5.8	10.9	3.6	30.0	48.9	6.6	8.5
202	1.0DL+1.0LR	236.9	-11.7	5.0	14.4	8.0	35.6	22.3	8.1	6.6
203	1.0DL+0.75LL+0.75LR+0.75TL	236.3	-13.5	5.6	11.6	4.6	31.6	38.5	7.0	8.1
204	1.0DL+1.0WLX1	239.8	-21.5	17.4	15.0	8.4	10.3	21.5	4.5	6.4
205	1.0DL+1.0WLX2	239.7	-22.8	17.9	14.9	8.3	10.1	21.6	4.2	6.4
206	1.0DL+1.0WLZ1	233.9	-9.3	4.2	9.8	3.4	41.6	51.4	10.1	9.5
207	1.0DL+1.0WLZ2	233.8	-10.7	4.7	9.8	3.4	37.7	51.5	8.8	9.5
208	1.0DL+0.75LL+0.75LR+0.75WLX1	246.3	-18.9	14.3	14.0	7.9	12.9	23.3	5.2	7.0
209	1.0DL+0.75LL+0.75LR+0.75WLX2	246.2	-20.0	14.6	14.0	7.9	12.6	23.4	4.9	7.0
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	241.9	-9.8	4.4	10.1	4.2	41.5	43.0	9.9	9.5
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	241.8	-10.8	4.7	10.1	4.2	38.6	43.0	8.9	9.5
212	0.6DL+1.0WLX1	147.4	-16.8	15.5	9.1	5.1	7.2	21.6	3.5	6.5
213	0.6DL+1.0WLX2	147.2	-18.2	15.9	9.0	5.1	6.9	21.7	3.2	6.5
214	0.6DL+1.0WLZ1	141.5	-4.6	2.2	3.9	0.2	47.5	617.0	12.2	14.4
215	0.6DL+1.0WLZ2	141.4	-6.0	2.7	3.9	0.2	39.8	646.5	9.4	14.4
						Min.=	6.9	21.5	3.2	6.4
							OK	OK	OK	OK

1.5

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Load Combination & Calculation of Base pressure

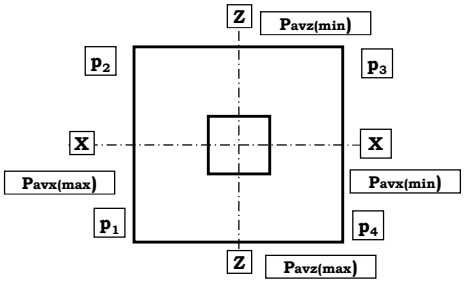
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 10 kN/m^2

Net SBC of soil for DL, LL= 350 kN/m^2 For WL/SL case = 437.5 kN/m^2

Gross SBC of soil for DL, LL= 377 kN/m^2 For WL/SL case = 464.5 kN/m^2



Allowable Stress Design																	
Load Case	Load Combination	Design Load/moments			Base Pressure (kN/m^2)												
		V (kN)	Mz (kN-m)	Mx (kN-m)	V/ (Lx.Lz)	6.Mz/ $(Lx)^2$.Lz	6.Mx/ Lx.$(Lz)^2$	p_1 (p_{max})	p_2	p_4	p_3 (p_{min})	(p_{max}-gh)	$p_{avx(max)}$	$p_{avx(min)}$	$p_{avz(max)}$	$p_{avz(min)}$	Remarks
201	1.0DL+1.0LL+1.0TL	232.33	5.82	3.57	103.26	10.34	6.34	119.94	107.26	99.25	86.58	92.94	113.60	92.92	109.60	96.92	O.K
202	1.0DL+1.0LR	236.86	4.99	7.97	105.27	8.87	14.17	128.31	99.97	110.58	82.23	101.31	114.14	96.41	119.45	91.10	O.K
203	1.0DL+0.75LL+0.75LR+0.75TL	236.33	5.62	4.60	105.04	9.99	8.18	123.20	106.85	103.23	86.88	96.20	115.03	95.06	113.22	96.87	O.K
204	1.0DL+1.0WLX1	239.80	17.44	8.35	106.58	31.01	14.85	152.43	122.74	90.42	60.73	125.43	137.59	75.58	121.43	91.74	O.K
205	1.0DL+1.0WLX2	239.70	17.88	8.33	106.53	31.79	14.80	153.12	123.52	89.54	59.94	126.12	138.32	74.74	121.33	91.73	O.K
206	1.0DL+1.0WLZ1	233.95	4.22	3.42	103.98	7.51	6.07	117.55	105.41	102.54	90.40	90.55	111.48	96.47	110.05	97.91	O.K
207	1.0DL+1.0WLZ2	233.82	4.66	3.41	103.92	8.28	6.06	118.25	106.14	101.70	89.59	91.25	112.20	95.65	109.98	97.87	O.K
208	1.0DL+0.75LL+0.75LR+0.75WLX1	246.26	14.28	7.92	109.45	25.39	14.08	148.92	120.76	98.14	69.98	121.92	134.84	84.06	123.53	95.37	O.K
209	1.0DL+0.75LL+0.75LR+0.75WLX2	246.18	14.61	7.90	109.41	25.98	14.05	149.44	121.35	97.48	69.39	122.44	135.40	83.44	123.46	95.37	O.K
210	1.0DL+0.75LL+0.75LR+0.75WLZ1	241.87	4.37	4.22	107.50	7.77	7.50	122.76	107.76	107.23	92.23	95.76	115.26	99.73	115.00	100.00	O.K
211	1.0DL+0.75LL+0.75LR+0.75WLZ2	241.78	4.69	4.21	107.46	8.34	7.49	123.29	108.31	106.60	91.62	96.29	115.80	99.11	114.95	99.97	O.K
212	0.6DL+1.0WLX1	147.35	15.45	5.11	65.49	27.47	9.08	102.04	83.88	47.10	28.94	75.04	92.96	38.02	74.57	56.41	O.K
213	0.6DL+1.0WLX2	147.25	15.89	5.08	65.44	28.25	9.03	102.73	84.66	46.22	28.15	75.73	93.70	37.19	74.48	56.41	O.K
214	0.6DL+1.0WLZ1	141.50	2.23	0.17	62.89	3.97	0.31	67.16	66.55	59.22	58.61	40.16	66.86	58.92	63.19	62.58	O.K
215	0.6DL+1.0WLZ2	141.37	2.67	0.16	62.83	4.74	0.29	67.86	67.28	58.38	57.80	40.86	67.57	58.09	63.12	62.54	O.K

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Ultimate Conditions

101 1.4DL	323.57	6.96	11.35	143.81	12.37	20.18	176.36	136.01	151.61	111.26	149.36	156.19	131.44	163.99	123.64
102 1.4DL+1.7LL+0.5LR	336.24	6.85	10.67	149.44	12.17	18.98	180.59	142.64	156.24	118.29	153.59	161.62	137.27	168.42	130.47
103 1.2DL+1.6LR+0.8WLX1	293.48	15.97	9.71	130.43	28.39	17.26	176.08	141.56	119.31	84.79	149.08	158.82	102.05	147.70	113.18
104 1.2DL+1.6LR+0.8WLX2	293.39	16.32	9.69	130.40	29.01	17.22	176.63	142.19	118.61	84.16	149.63	159.41	101.39	147.62	113.18
105 1.2DL+1.6LR+0.8WLZ1	288.79	5.39	5.76	128.35	9.59	10.24	148.18	127.70	129.00	108.53	121.18	137.94	118.77	138.59	118.12
106 1.2DL+1.6LR+0.8WLZ2	288.69	5.74	5.75	128.31	10.20	10.23	148.74	128.28	128.33	107.88	121.74	138.51	118.11	138.54	118.08
107 1.2DL+1.6LR+1.0LL	292.30	5.92	9.16	129.91	10.53	16.28	156.71	124.16	135.66	103.11	129.71	140.44	119.39	146.19	113.64
108 1.2DL+1.0LL+1.6WLX1+0.5LR	299.87	25.86	9.69	133.28	45.96	17.23	196.47	162.01	104.54	70.08	169.47	179.24	87.31	150.51	116.05
109 1.2DL+1.0LL+1.6WLX1+0.5LR	299.70	26.56	9.65	133.20	47.22	17.16	197.58	163.26	103.14	68.82	170.58	180.42	85.98	150.36	116.04
110 1.2DL+1.0LL+1.6WLX1+0.5LR	290.50	4.70	1.80	129.11	8.36	3.19	140.67	134.28	123.94	117.56	113.67	137.48	120.75	132.31	125.92
111 1.2DL+1.0LL+1.6WLX1+0.5LR	290.30	5.40	1.78	129.02	9.59	3.17	141.79	135.45	122.60	116.26	114.79	138.62	119.43	132.20	125.86
112 1.2DL+1.0LL	283.11	5.90	9.37	125.83	10.48	16.66	152.97	119.65	132.01	98.69	125.97	136.31	115.35	142.49	109.17
113 0.9DL+1.6WLX1	221.90	24.43	7.69	98.62	43.42	13.66	155.71	128.38	68.86	41.53	128.71	142.05	55.20	112.29	84.96
114 0.9DL+1.6WLX2	221.73	25.13	7.65	98.55	44.68	13.59	156.81	129.63	67.46	40.28	129.81	143.22	53.87	112.14	84.96
115 0.9DL+1.6WLZ1	212.53	3.28	-0.21	94.46	5.82	0.37	100.65	99.91	89.01	88.26	73.65	100.28	88.64	94.83	94.09
116 0.9DL+1.6WLZ2	212.33	3.97	-0.22	94.37	7.05	0.40	101.82	101.02	87.71	86.92	74.82	101.42	87.32	94.77	93.97
117 1.4DL+ 1.4TL+1.7LL+0.5LR	329.86	8.13	4.82	146.60	14.45	8.56	169.62	152.50	140.71	123.59	142.62	161.06	132.15	155.17	138.05
Max :	336	27	11												
Min :	141	2	0												

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL cases) = 92.94 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL cases) = 102.94 kN/m²

Max pressure at base of foundation without surcharge excluding weight of soil (DL+LL+SL/WL cases) = 126.12 kN/m²

Max pressure at base of foundation including surcharge excluding weight of soil (DL+LL+WL/SL cases) = 136.12 kN/m²

Sample calculation of values of base pressure:

Load case : 202 1.0DL+1.0LR

Maximum pressure, p_1 (pmax) = 105.27+8.87+14.17= 128.31 kN/m²

p_2= 105.27+8.87-14.17= 99.97 kN/m²

p_3= 105.27-(8.87+14.17)= 82.23 kN/m²

p_4= 105.27-8.87+14.17= 110.58 kN/m²

Maxm pressure excluding weight of soil (p_{max}-gh)= 128.31-1.5*18= 101.31 kN/m²

Maximum average Pressure along X, $p_{avx(max)}$ = (128.31+99.97)/2= 114.14 kN/m²

Minimum average Pressure along X, $p_{avx(minx)}$ = (82.23+110.58)/2= 96.41 kN/m²

Maximum average Pressure along Y, $p_{avy(max)}$ = (128.31+110.58)/2= 119.45 kN/m²

Minimum average Pressure along Y, $p_{avy(minx)}$ = (82.23+99.97)/2= 91.10 kN/m²

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Calculation of Bending Moment and Shear Force in X direction :

p_{1x} =pressure at the face of column at section 1x-1x in kN/m^2.

p_{2x} =pressure at the face of column at section 2x-2x in kN/m^2.

p_{3x} =pressure at a distance 'd' from the face of column at section 3x-3x in kN/m^2.

BM_{1x} =Moment at the face of column at section 1x-1x in kN-m/m.

BM_{2x} =Moment at the face of column 2x-2x in kN-m/m.

SF_{3x} = Oneway shear force at a distance 'd' from the face of column at section 3x-3x in kN/m.

SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Punching perimeter= ((0.4+2*0.5*0.326)+(0.4+2*0.5*0.326))*2

= 2.904 m

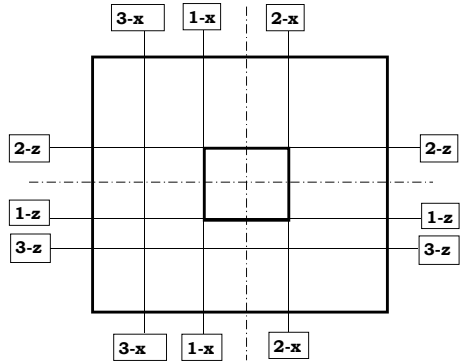
Punching Area= ((0.4+2*0.5*0.326)*(0.4+2*0.5*0.326))

= 0.53 m^2 Punching shear stress to be checked

Downward pressure due to self weight of raft and soil = (0.4*24+1.1*18)

= 29.40 kN/m^2

Area of footing, A_f= 2.25 m^2



Load Case	Load Combination	p_{1x} kN/m2	p_{2x} kN/m2	p_{3x} kN/m2	B.M$_{1x}$ kNm	B.M$_{2x}$ kNm	S.F$_{3x}$ kN	S.F$_{(punching)}$ kN
Ultimate Conditions								
101	1.4DL	147.12	140.52	152.76	18.72	15.89	26.02	196.79
102	1.4DL+1.7LL+0.5LR	152.69	146.20	158.24	19.55	16.77	27.15	206.47
103	1.2DL+1.6LR+0.8WLX1	138.00	122.87	150.95	18.53	12.04	26.10	173.78
104	1.2DL+1.6LR+0.8WLX2	138.14	122.66	151.36	18.59	11.96	26.21	173.71
105	1.2DL+1.6LR+0.8WLZ1	130.91	125.80	135.28	16.06	13.87	22.30	170.20
106	1.2DL+1.6LR+0.8WLZ2	131.03	125.59	135.68	16.13	13.79	22.40	170.12
107	1.2DL+1.6LR+1.0LL	132.72	127.11	137.52	16.41	14.00	22.79	172.88
108	1.2DL+1.0LL+1.6WLX1+0.5LR	145.53	121.02	166.49	20.96	10.46	29.84	178.67
109	1.2DL+1.0LL+1.6WLX1+0.5LR	145.79	120.61	167.32	21.10	10.30	30.05	178.54
110	1.2DL+1.0LL+1.6WLX1+0.5LR	131.35	126.88	135.16	16.04	14.13	22.24	171.50
111	1.2DL+1.0LL+1.6WLX1+0.5LR	131.58	126.47	135.96	16.16	13.97	22.44	171.35

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112 1.2DL+1.0LL	128.62	123.04	133.40	15.78	13.39	21.94	165.86
113 0.9DL+1.6WLX1	110.21	87.05	130.01	15.43	5.51	22.18	119.06
114 0.9DL+1.6WLX2	110.46	86.63	130.83	15.56	5.35	22.39	118.93
115 0.9DL+1.6WLZ1	96.01	92.91	98.67	10.51	9.18	14.58	111.90
116 0.9DL+1.6WLZ2	96.25	92.49	99.46	10.63	9.02	14.78	111.74
117 1.4DL+ 1.4TL+1.7LL+0.5LR	150.46	142.75	157.05	19.38	16.08	26.97	201.59

Sample calculation of BM & SF

Load case : 101 1.4DL

p_{1x} = 156.19-(156.19-131.44)*0.55/1.5 = 147.12 kN/m^2 Overhang = 0.55 m

p_{2x} = 131.44+(156.19-131.44)*0.55/1.5 = 140.52 kN/m^2 Effective depth = 0.342 m

p_{3x} = 156.19-(156.19-131.44)*(0.55-0.342)/1.5 = 152.76 kN/m^2

BM_{1x} = (147.12-29.4)*0.55*0.55/2+0.5*(156.19-147.12)*0.55*0.55*0.67 = 18.72 kN-m/m

BM_{2x} = (140.52-29.4)*0.55*0.55/2-0.5*(-131.44+140.52)*0.55*0.55*0.67 = 15.89 kN-m/m

SF_{3x} = (152.76-29.4)*(0.55-0.342)+0.5*(156.19-152.76)*(0.55-0.342) = 26.02 kN/m

$S.F_{(punching)}$ = (2.25-0.53)*(143.81-29.4) = 196.79 kN

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	21.10	0.18	0.24	0.001	196.62	615.60	16ϕT	326.61	300	200	1005.31
At top	0.00	0.00	0.00	0.000	0.00	205.20	10ϕT	382.75	300	200	392.70

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, $(\phi\ Vc)/bd$ = 0.740 N/mm^2

Shear at a distance 'd' from face of column, Vu= 30.05 kN/m

Shear stress , τ_v= (30.05*1000)/(1000*342)

= 0.09 N/mm^2 **O.K.**

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Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu=	206.47 kN	
Strength of Concrete in shear Vc1 =	2937.83 kN	**Table 22.6.5.2_ (a)**
Strength of Concrete in shear Vc2 =	2132.56 kN	**Table 22.6.5.2_ (b)**
Strength of Concrete in shear Vc3 =	1958.55 kN	**Table 22.6.5.2_ (c)**
Design Vc=	1958.55 kN	
Allowable shear stress, $(\phi\ \mathrm{Vc})/\mathrm{bd}$ =	1.48 $\mathrm{N/mm^2}$	
Design punching shear stress, τ =	(206.47*1000)/(2904*342)	
=	0.21 $\mathrm{N/mm^2}$	**No shear reinforcement reqd.**

Calculation of Bending Moment and Shear Force in Z direction :

p1z =pressure at the face of column at section 1z-1z in $\mathrm{kN/m^2}$.
p2z =pressure at the face of column at section 2z-2z in $\mathrm{kN/m^2}$.
p3z =pressure at a distance d from the face of column at section 3z-3z in $\mathrm{kN/m^2}$.
B.M1z =Moment at the face of column at section 1z-1z in kN-m/m.
B.M2z =Moment at the face of column at section 2z-2z in kN-m/m.
S.F 3z = Oneway Shear Force at a distance d from the face of column at section 3z-3z in kN/m.
S.F (punching) = Punching Shear Force at a distance 0.5d from the face of column in kN/m.

Load Case	Load Combination	p_{1z} kN/m2	p_{2z} kN/m2	p_{3z} kN/m2	$B.M_{1z}$ kNm	$B.M_{2z}$ kNm	$S.F_{3z}$ kN
Ultimate Conditions							
101	1.4DL	149.20	138.44	157.96	19.62	14.99	29.47
102	1.4DL+1.7LL+0.5LR	154.51	144.39	162.75	20.33	15.98	30.51
103	1.2DL+1.6LR+0.8WLX1	135.04	125.84	142.55	17.26	13.30	25.92
104	1.2DL+1.6LR+0.8WLX2	134.99	125.81	142.48	17.25	13.30	25.91
105	1.2DL+1.6LR+0.8WLZ1	131.08	125.63	135.53	16.14	13.79	24.12
106	1.2DL+1.6LR+0.8WLZ2	131.04	125.58	135.48	16.13	13.79	24.11
107	1.2DL+1.6LR+1.0LL	134.26	125.58	141.33	17.07	13.34	25.62
108	1.2DL+1.0LL+1.6WLX1+0.5LR	137.87	128.69	145.36	17.69	13.74	26.55
109	1.2DL+1.0LL+1.6WLX1+0.5LR	137.78	128.62	145.23	17.67	13.73	26.52
110	1.2DL+1.0LL+1.6WLX1+0.5LR	129.97	128.26	131.36	15.45	14.72	22.94
111	1.2DL+1.0LL+1.6WLX1+0.5LR	129.88	128.18	131.25	15.43	14.71	22.92

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112 1.2DL+1.0LL	130.27	121.39	137.51	16.50	12.68	24.77
113 0.9DL+1.6WLX1	102.27	94.98	108.21	12.04	8.90	18.11
114 0.9DL+1.6WLX2	102.17	94.93	108.08	12.02	8.90	18.08
115 0.9DL+1.6WLZ1	94.56	94.36	94.72	9.88	9.80	14.64
116 0.9DL+1.6WLZ2	94.48	94.26	94.65	9.87	9.78	14.63
117 1.4DL+ 1.4TL+1.7LL+0.5LR	148.89	144.33	152.61	18.71	16.75	27.89

Sample calculation of BM & SF

Load case : 101 1.4DL

p_{1z} = 163.99-(163.99-123.64)*0.55/1.5= 149.20 kN/m^2

p_{2z} = 123.64+(163.99-123.64)*0.55/1.5= 138.44 kN/m^2

p_{3z} = 163.99-(163.99-123.64)*(0.55-0.326)/1.5= 157.96 kN/m^2

Overhang = 0.55 m

Effective depth = 0.326 m

BM_{1z} = (149.20-29.4)*0.55*0.55/2+0.5*(163.99-149.20)*0.55*0.55*0.67 = 19.62 kN-m/m

BM_{2z} = (138.44-29.4)*0.55*0.55/2-0.5*(-123.64+138.44)*0.55*0.55*0.67 = 14.99 kN-m/m

SF_{3z} = (157.96-29.4)*(0.55-0.326)+0.5*(163.99-157.96)*(0.55-0.326) = 29.47 kN/m

Location	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
At bottom	20.33	0.191	0.255	0.0006	208.61	586.80	16ϕT	342.64	300	200	1005.3
At top	0.00	0.000	0.000	0.0000	0.00	195.60	10ϕT	401.53	300	200	392.7

Check against one-way shear force at a distance d from face of column:

Allowable shear stress, (ϕ Vc) = 0.740 N/mm2

Shear at a distance 'd' from face of column, Vu= 30.51 kN/m

Shear stress , τ_v= (30.51*1000)/(1000*326)

= 0.09 N/mm2 **O.K.**

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :				Along Z Direction :			
At Bottom	16ϕT	200	c/c	At Bottom	16ϕT	200	c/c
At Top	10ϕT	200	c/c	At Top	10ϕT	200	c/c

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REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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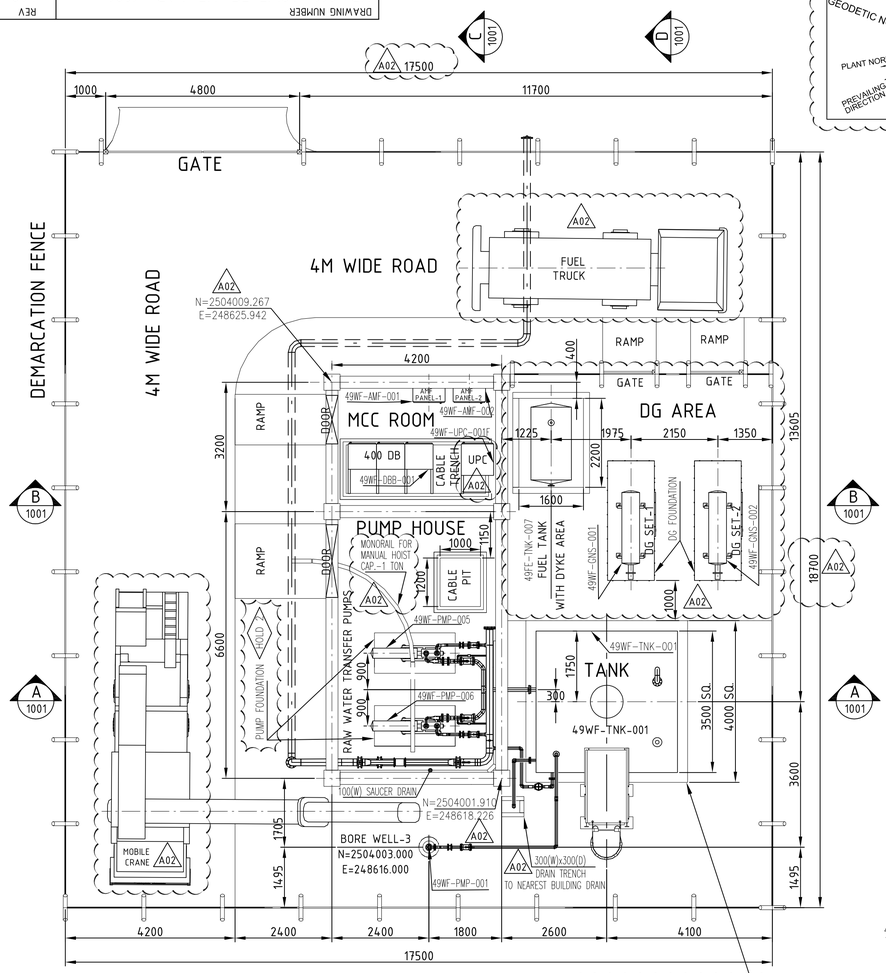
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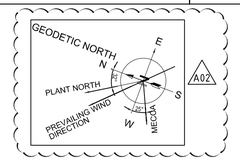
ATTACHMENT 1 INPUT GA DRAWINGS

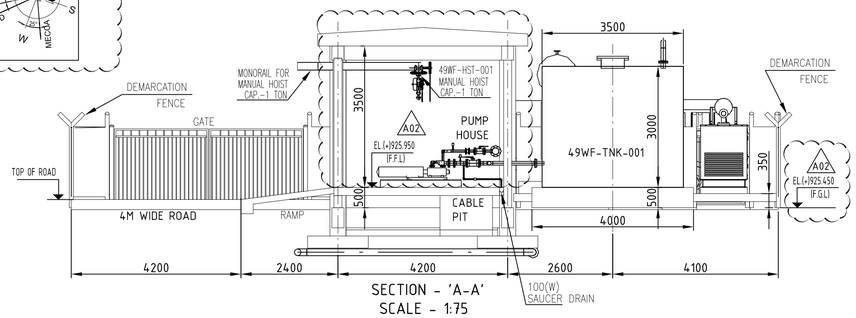
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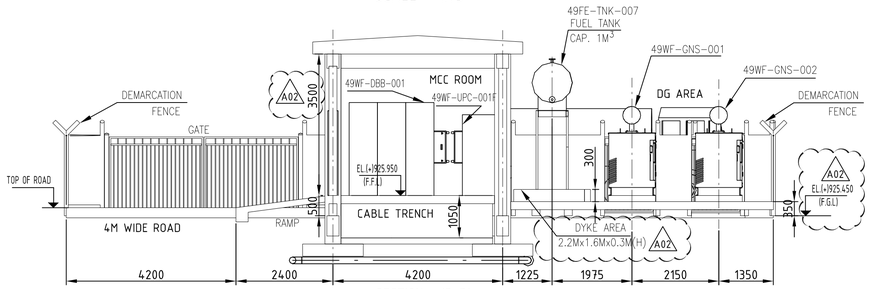
DRAWING NUMBER MD-216-49WF-EG-ME-LAD-1001 REV A02



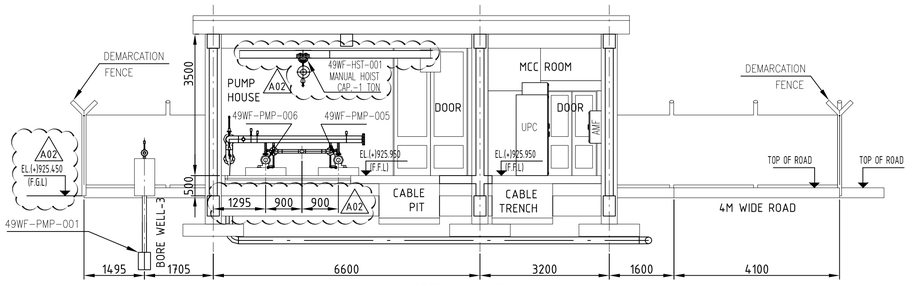




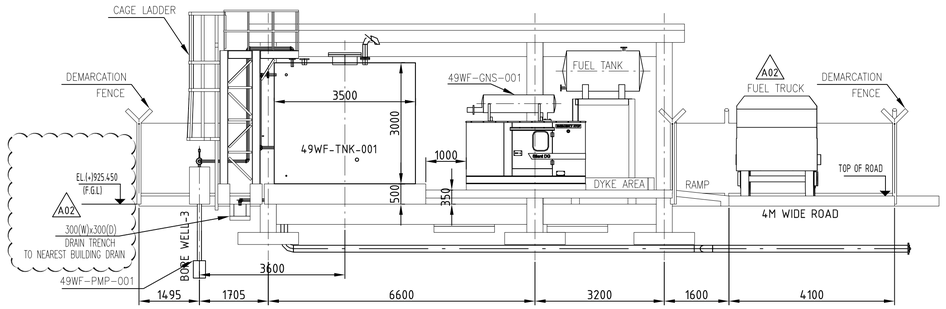
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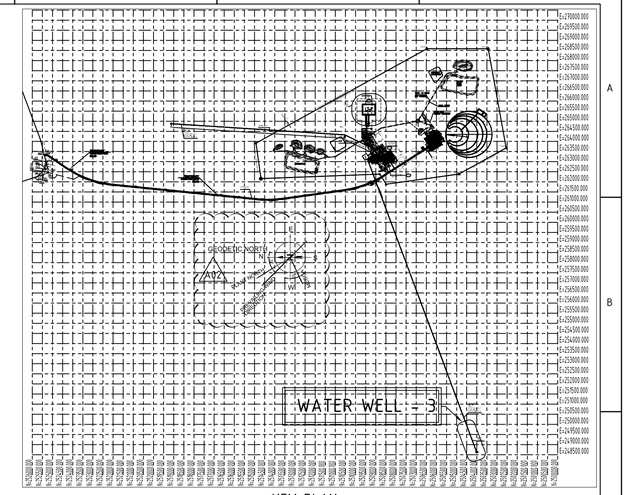
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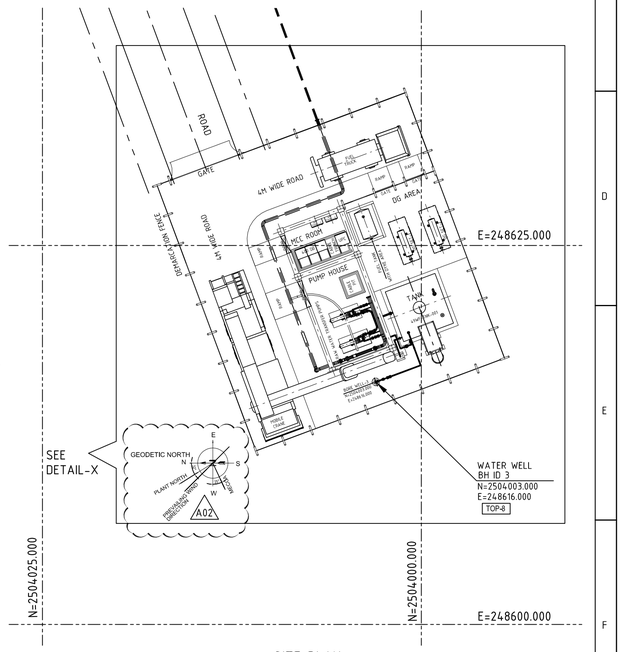
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SCALE - 1:75



SECTION - 'D-D'
SCALE - 1:75



KEY PLAN
SCALE - 1:150000



SITE PLAN
SCALE - 1:200

SL. No.	EQUIPMENT	TAG	QTY.
1.	BORE WELL WATER PUMP1	49WF-PMP-001	1 No.
2.	RAW WATER TRANSFER PUMP1	49WF-PMP-005	1 No.
3.	RAW WATER TRANSFER PUMP2	49WF-PMP-006	1 No.
4.	BORE WELL WATER STORAGE TANK	49WF-TNK-001	1 No.
5.	DG SET-1	49WF-GNS-001	1 No.
6.	DG SET-2	49WF-GNS-002	1 No.
7.	DBB	49WF-DBB-001	1 No.
8.	AMF PANEL-1	49WF-AMF-001	1 No.
9.	AMF PANEL-2	49WF-AMF-002	1 No.
10.	FUEL TANK	49FE-TNK-007	1 No.
11.	UPC	49WF-UPC-001F	1 No.
12.	MANUAL HOIST (CAP. - 1 TON)	49WF-HST-001	1 No.

A02

ENLARGE DETAIL - X
(GA OF BORE WELL WATER PUMP STATION AT BH-3)
SCALE - 1:75

3500x3500x3000 HT. HDPE TANK WITH 4000x4000x500 HT. FOUNDATION
HOLD 1

NOTES:

1. ALL DIMENSIONS ARE IN MM. AND LEVELS ARE IN M.
2. ALL CO-ORDINATES ARE IN METERS.
3. CO-ORDINATES ARE ON WORLD GRID.
4. RAW WATER PUMPS & TANK FOUNDATION ARE UNDERHOLD AND WILL BE FINALISED AFTER GETTING EQUIPMENT DRAWING.
5. INSTRUMENT LOCATION SHALL BE INDICATED IN SEPARATE PIPING GA DRAWING. A02

ABBREVIATIONS / LEGENDS

FGL – FINISH GRADE LEVEL
FFL – FINISH FLOOR LEVEL
HT – HEIGHT
CAP – CAPACITY
ARRGT – ARRANGEMENT
PS – PUMP STATION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	07/12/2019	ISSUED FOR DETAIL DESIGN	GAN	ABS	SRS	KBJ	
		A01	30/08/2019	ISSUED FOR APPROVAL	GAN	ABS	SRS	KBJ	

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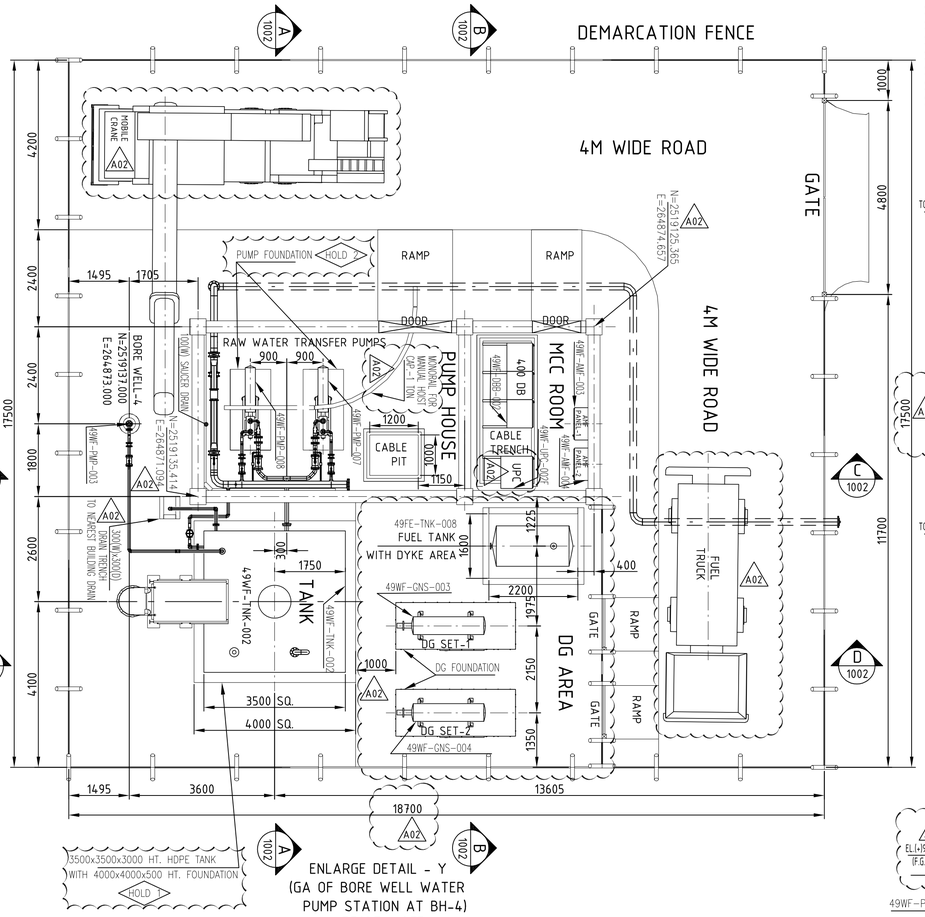
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		OU100660961	1

DRAWN	GAN	30/08/2019
CHECKED	ABS	30/08/2019
ENGINEER	SRS	30/08/2019
APPROVED	KBJ	30/08/2019
EMPLOYER		
SCALE	1:75, 200	A1

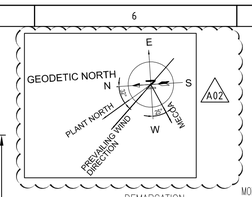
MANSOURAH - MASSARAH GOLD PROJECT
LAYOUT
POTABLE WATER SUPPLY EQUIPMENT ARRGT. FOR BOREWELL PS (BH3) A02

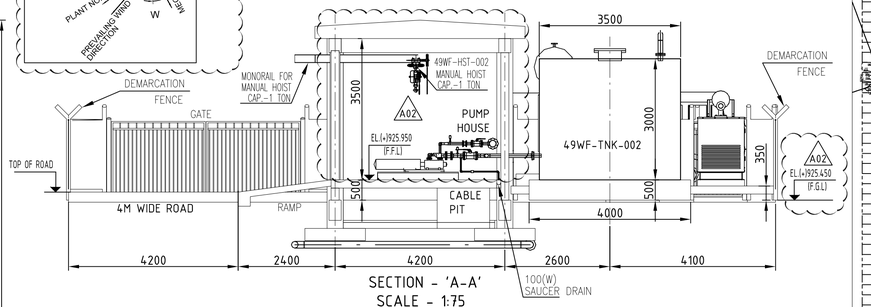
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MD-216-49WF-EG-ME-LAD-1001	A02

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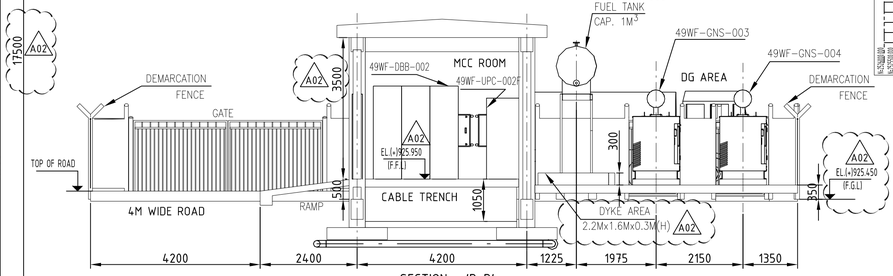


ENLARGE DETAIL - Y
(GA OF BORE WELL WATER PUMP STATION AT BH-4)
SCALE - 1:75

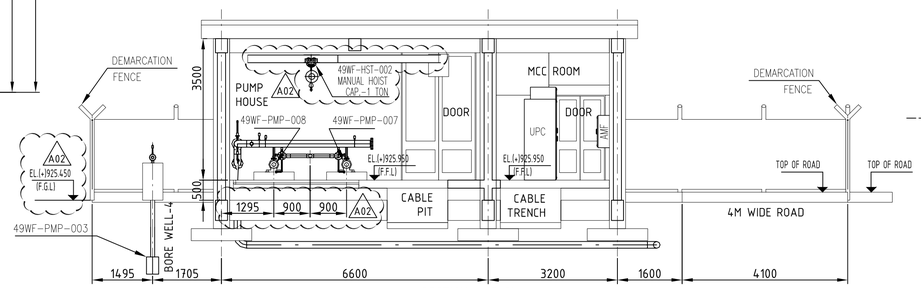




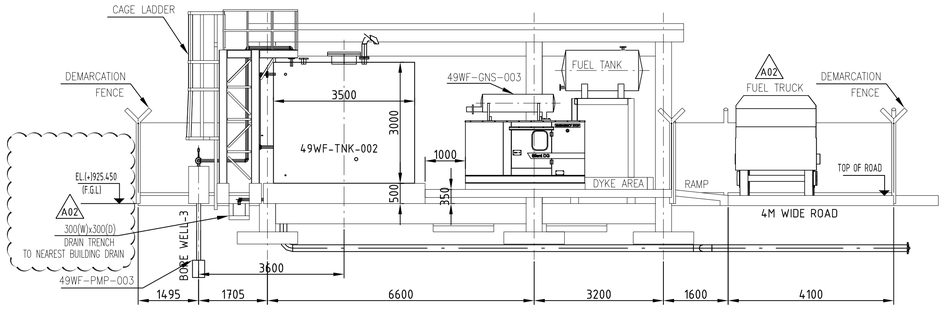
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SCALE - 1:75



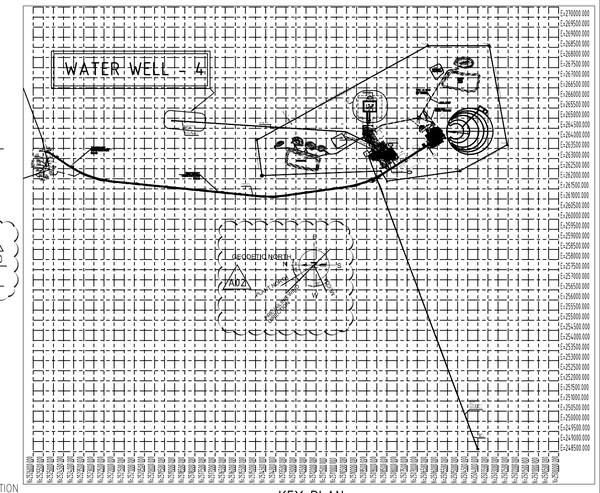
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SCALE - 1:75



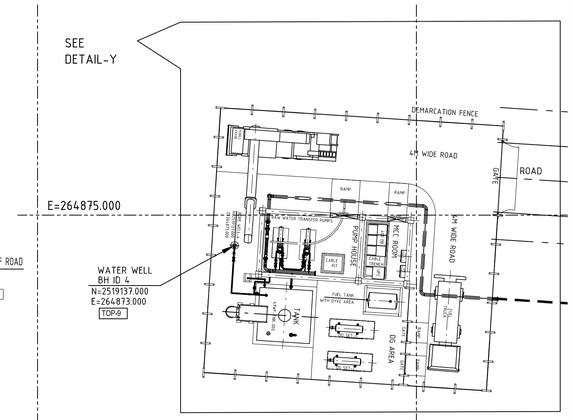
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SCALE - 1:75



SECTION - 'D-D'
SCALE - 1:75



KEY PLAN
SCALE - 1:150000



SITE PLAN
SCALE - 1:200

SL. No.	EQUIPMENT	TAG	QTY.
1.	BORE WELL WATER PUMP1	49WF-PMP-003	1 No.
2.	RAW WATER TRANSFER PUMP1	49WF-PMP-007	1 No.
3.	RAW WATER TRANSFER PUMP2	49WF-PMP-008	1 No.
4.	BORE WELL WATER STORAGE TANK	49WF-TNK-002	1 No.
5.	DG SET-1	49WF-GNS-003	1 No.
6.	DG SET-2	49WF-GNS-004	1 No.
7.	DBB	49WF-DBB-002	1 No.
8.	AMF PANEL-1	49WF-AMF-003	1 No.
9.	AMF PANEL-2	49WF-AMF-004	1 No.
10.	FUEL TANK	49FE-TNK-008	1 No.
11.	UPC	49WF-UPC-002F	1 No.
12.	MANUAL HOIST (CAP. - 1 TON)	49WF-HST-002	1 No.

A02

NOTES:

1. ALL DIMENSIONS ARE IN MM. AND LEVELS ARE IN M.
2. ALL CO-ORDINATES ARE IN METERS.
3. CO-ORDINATES ARE ON WORLD GRID.
4. RAW WATER PUMPS & TANK FOUNDATION ARE UNDERHOLD AND WILL BE FINALISED AFTER GETTING EQUIPMENT DRAWING.
5. INSTRUMENT LOCATION SHALL BE INDICATED IN SEPARATE PIPING GA DRAWING. A02

A02

ABBREVIATIONS / LEGENDS
FGL – FINISH GRADE LEVEL
FFL – FINISH FLOOR LEVEL
HT – HEIGHT
CAP – CAPACITY
ARRGT – ARRANGEMENT
PS – PUMP STATION

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A02	07/12/2019	ISSUED FOR DETAIL DESIGN	GAN	ABS	SRS	KBJ	
		A01	30/08/2019	ISSUED FOR APPROVAL	GAN	ABS	SRS	KBJ	

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OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
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CHECKED	ABS	30/08/2019	LAYOUT
ENGINEER	SRS	30/08/2019	POTABLE WATER SUPPLY A02
APPROVED	KBJ	30/08/2019	EQUIPMENT ARRGT. FOR BOREWELL PS (BH4)
EMPLOYER			DRAWING NUMBER / REV
SCALE	1:75, 200	A1	MD-216-49WF-EG-ME-LAD-1002 / A02

ATTACHMENT 2 ISOMETRIC VIEW OF STAAD MODEL FOR PUMP STATION

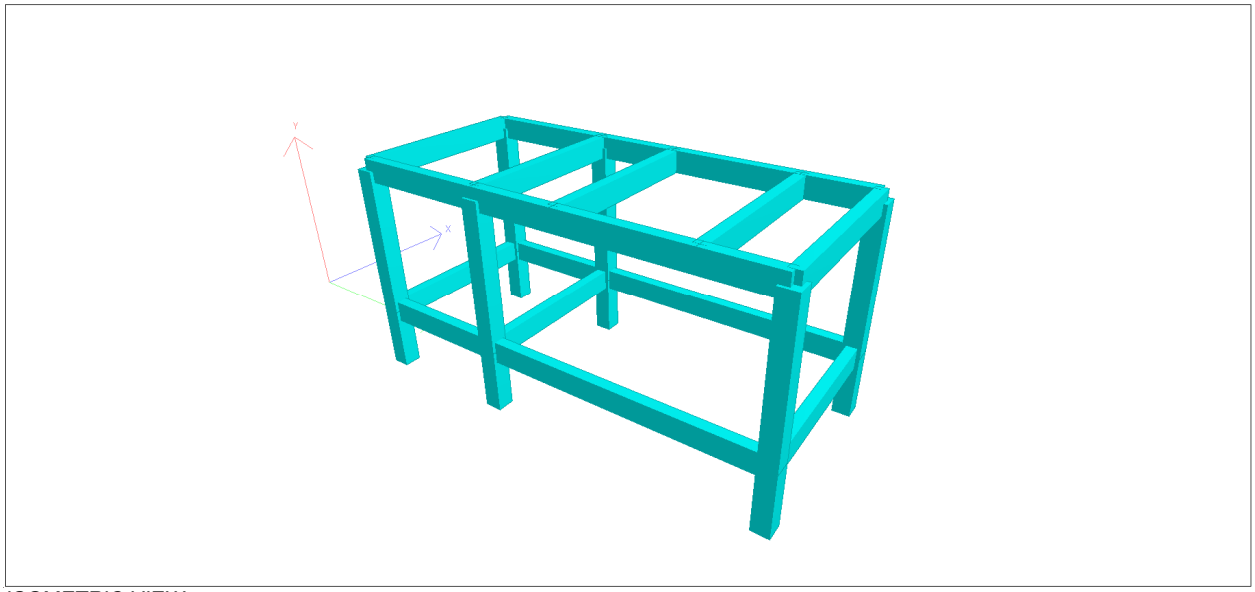
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STAAD.Pro Report

To: From:

Copy to: Date: 28/01/2020 14:52:00 Ref: ca/ Document1



ISOMETRIC VIEW

ATTACHMENT 3 STAAD INPUT BORE WELL PUMP STATION

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```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 15.01.2020
ENGINEER NAME SOR
CHECKER NAME SJM
APPROVED NAME AG
JOB CLIENT MAA'DEN
JOB REV 0
JOB NAME BORE WELL PUMP STATION
JOB NO O19045
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0.122 0 2.585; 2 4.32201 0 2.585; 4 0.122 3.49961 2.585;
5 4.32201 3.49961 2.585; 7 0.122 0 5.78501; 10 0.122 3.49961 5.78501;
23 0.122 -1.5 2.585; 24 4.32201 -1.5 2.585; 26 0.122 -1.5 5.78501;
34 0.122 0 12.385; 35 4.32201 0 12.385; 36 0.122 3.49961 12.385;
37 4.32201 3.49961 12.385; 38 0.122 -1.5 12.385; 39 4.32201 -1.5 12.385;
79 4.322 0 5.78501; 80 4.322 3.49961 5.78501; 81 4.322 -1.5 5.78501;
82 0.122 3.49961 7.78501; 83 4.322 3.49961 7.78501; 84 0.122 3.49961 10.785;
85 4.32201 3.49961 10.785;
MEMBER INCIDENCES
1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24;
34 7 26; 488 7 34; 490 10 82; 497 36 37; 498 34 36; 499 35 37; 500 35 34;
501 34 38; 502 35 39; 583 79 80; 584 79 2; 585 79 81; 586 79 35; 587 5 80;
588 80 83; 589 7 79; 590 10 80; 591 82 84; 592 83 85; 593 82 83; 594 84 36;
595 85 37; 596 84 85;
START GROUP DEFINITION
MEMBER
_COLUMN_C1 3 4 8 31 32 34 498 499 501 502 583 585
_RB_1 11 490 587 588 591 592 594 595
_RB_2 590
_GB_1 25 30 488 500 584 586 589
_RB-3 593 596
_RB_4 1 497
END GROUP DEFINITION
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17184e+007
POISSON 0.17
DENSITY 23.6158
ALPHA 5e-006
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27578.9
END DEFINE MATERIAL
CONSTANTS
MATERIAL CONCRETE ALL
MEMBER PROPERTY AMERICAN
3 4 8 31 32 34 498 499 501 502 583 585 PRIS YD 0.4 ZD 0.4
1 497 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
25 30 488 500 584 586 589 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
11 490 587 588 591 592 594 595 PRIS YD 0.5 ZD 0.25
MEMBER PROPERTY AMERICAN
590 593 596 PRIS YD 0.45 ZD 0.25
SUPPORTS
23 24 26 38 39 81 FIXED
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LOAD 1 LOADTYPE Dead  TITLE DL
SELFWEIGHT Y -1
MEMBER LOAD
25 30 488 500 584 586 589 UNI GY -12
11 490 587 588 591 592 594 595 UNI GY -13.45
11 490 591 594 UMOM GZ 1
587 588 592 595 UMOM GZ -1
1 UMOM GX -1
497 UMOM GX 1
*************************************************************************************************************
****
LOAD 2 LOADTYPE Live  TITLE LL
MEMBER LOAD
* LOAD DUE TO MONORAIL
593 596 CON GY -10 2.1
490 CON GY -10 1
*************************************************************************************************************
****
LOAD 3 LOADTYPE Roof Live  TITLE LR
MEMBER LOAD
11 490 587 588 591 592 594 595 UNI GY -1.05
11 490 591 594 UMOM GZ 0.1
```



```
*************************************************************************************************************
****
LOAD 4 LOADTYPE Wind  TITLE WIND ALONG X DIR. ( CPI = + 0.18) WLX1
MEMBER LOAD
3 UNI GX 0.9
8 UNI GX 2.74
498 UNI GX 1.85
4 UNI GX 1.28
583 UNI GX 3.92
499 UNI GX 2.64
498 499 UNI GZ 1.74
3 4 UNI GZ -1.74
*************************************************************************************************************
****
LOAD 5 LOADTYPE Wind  TITLE WIND ALONG X DIR. ( CPI = - 0.18) WLX2
MEMBER LOAD
3 UNI GX 1.65
8 UNI GX 5.05
498 UNI GX 3.4
4 UNI GX 0.53
583 UNI GX 1.62
499 UNI GX 1.09
498 499 UNI GZ 0.74
3 4 UNI GZ -0.74
*************************************************************************************************************
****
LOAD 6 LOADTYPE Wind  TITLE WIND ALONG Z DIR. ( CPI = + 0.18) WLZ1
MEMBER LOAD
3 UNI GX -1.33
8 UNI GX -4.07
498 UNI GX -2.74
4 UNI GX 1.33
583 UNI GX 4.07
499 UNI GX 2.74
498 499 UNI GZ 1.68
3 4 UNI GZ 1.18
*************************************************************************************************************
****
LOAD 7 LOADTYPE Wind  TITLE WIND ALONG Z DIR. ( CPI = - 0.18) WLZ2
MEMBER LOAD
3 UNI GX -0.56
8 UNI GX -1.72
498 UNI GX -1.16
4 UNI GX 0.56
583 UNI GX 1.72
499 UNI GX 1.16
498 499 UNI GZ 0.69
3 4 UNI GZ 2.16
*************************************************************************************************************
****
LOAD 8 LOADTYPE Temperature  TITLE TEMPERATURE LOAD ( TL )
TEMPERATURE LOAD
3 4 8 498 499 583 TEMP 25
1 11 490 497 587 588 590 TO 596 TEMP 0 25
3 4 8 498 499 583 TEMP 0 0 25
*************************************************************************************************************
****
* FACTORED LOAD COMBINATIONS
LOAD COMB 101 1.4DL
1 1.4
LOAD COMB 102 1.4DL+1.7LL+0.5LR
1 1.4 2 1.7 3 0.5
LOAD COMB 103 1.2DL+1.6LR+0.8WLX1
1 1.2 3 1.6 4 0.8
LOAD COMB 104 1.2DL+1.6LR+0.8WLX2
1 1.2 3 1.6 5 0.8
LOAD COMB 105 1.2DL+1.6LR+0.8WLZ1
1 1.2 3 1.6 6 0.8
LOAD COMB 106 1.2DL+1.6LR+0.8WLZ2
1 1.2 3 1.6 7 0.8
LOAD COMB 107 1.2DL+1.6LR+1.0LL
1 1.2 2 1.0 3 1.6
LOAD COMB 108 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 4 1.6
LOAD COMB 109 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 5 1.6
LOAD COMB 110 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 6 1.6
LOAD COMB 111 1.2DL+1.0LL+1.6WLX1+0.5LR
1 1.2 2 1.0 3 0.5 7 1.6
LOAD COMB 112 1.2DL+1.0LL
1 1.2 2 1.0
LOAD COMB 113 0.9DL+1.6WLX1
1 0.9 4 1.6
```

```
LOAD COMB 117 1.4DL+ 1.4TL+1.7LL+0.5LR
1 1.4 2 1.7 3 0.5 8 1.4
**********************************************************************************************************
****
* UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
LOAD COMB 201 1.0DL+1.0LL+1.0TL
1 1.0 2 1.0 8 1.0
LOAD COMB 202 1.0DL+1.0LR
1 1.0 3 1.0
LOAD COMB 203 1.0DL+0.75LL+0.75LR+0.75TL
1 1.0 2 0.75 3 0.75 8 0.75
LOAD COMB 204 1.0DL+1.0WLX1
1 1.0 4 1.0
LOAD COMB 205 1.0DL+1.0WLX2
1 1.0 5 1.0
LOAD COMB 206 1.0DL+1.0WLZ1
1 1.0 6 1.0
LOAD COMB 207 1.0DL+1.0WLZ2
1 1.0 7 1.0
LOAD COMB 208 1.0DL+0.75LL+0.75LR+0.75WLX1
1 1.0 2 0.75 3 0.75 4 0.75
LOAD COMB 209 1.0DL+0.75LL+0.75LR+0.75WLX2
1 1.0 2 0.75 3 0.75 5 0.75
LOAD COMB 210 1.0DL+0.75LL+0.75LR+0.75WLZ1
1 1.0 2 0.75 3 0.75 6 0.75
LOAD COMB 211 1.0DL+0.75LL+0.75LR+0.75WLZ2
1 1.0 2 0.75 3 0.75 7 0.75
LOAD COMB 212 0.6DL+1.0WLX1
1 0.6 4 1.0
LOAD COMB 213 0.6DL+1.0WLX2
1 0.6 5 1.0
LOAD COMB 214 0.6DL+1.0WLZ1
1 0.6 6 1.0
LOAD COMB 215 0.6DL+1.0WLZ2
1 0.6 7 1.0
**********************************************************************************************************
****
PERFORM ANALYSIS
LOAD LIST 101 TO 117
START CONCRETE DESIGN
CODE ACI
TRACK 2 ALL
FC 35000 ALL
CLB 0.04 ALL
CLS 0.04 ALL
CLT 0.04 ALL
MAXMAIN 20 MEMB 3 4 8 31 32 34 498 499 501 502 583 585
MINMAIN 20 MEMB 3 4 8 31 32 34 498 499 501 502 583 585
MAXMAIN 20 MEMB 1 11 490 497 587 588 590 TO 596
MINMAIN 20 MEMB 1 11 490 497 587 588 590 TO 596
MAXMAIN 16 MEMB 11 490 587 588 591 592 594 595
MINMAIN 16 MEMB 11 490 587 588 591 592 594 595
MAXMAIN 16 MEMB 25 30 488 500 584 586 589
MINMAIN 16 MEMB 25 30 488 500 584 586 589
MINSEC 10 ALL
DESIGN BEAM 1 11 25 30 488 490 497 500 584 586 TO 596
DESIGN COLUMN 3 4 8 31 32 34 498 499 501 502 583 585
END CONCRETE DESIGN
FINISH
```

ATTACHMENT 4 STAAD OUTPUT BORE WELL PUMP STATION

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```
**************************************************
*                                                *
*          STAAD.Pro V8i SELECTseries5           *
*          Version  20.07.10.64                  *
*          Proprietary Program of                *
*          Bentley Systems, Inc.                 *
*          Date=    JAN 29, 2020                 *
*          Time=    12:28:38                     *
*                                                *
*     USER ID: LNT ECC DIV                       *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: BOREWELL PUMP STATION.STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 15.01.2020
    4. ENGINEER NAME SOR
    5. CHECKER NAME SJM
    6. APPROVED NAME AG
    7. JOB CLIENT MAA'DEN
    8. JOB REV 0
    9. JOB NAME BORE WELL PUMP STATION
   10. JOB NO O19045
   11. END JOB INFORMATION
   12. INPUT WIDTH 79
   13. UNIT METER KN
   14. JOINT COORDINATES
   15. 1 0.122 0 2.585; 2 4.32201 0 2.585; 4 0.122 3.49961 2.585
   16. 5 4.32201 3.49961 2.585; 7 0.122 0 5.78501; 10 0.122 3.49961 5.78501
   17. 23 0.122 -1.5 2.585; 24 4.32201 -1.5 2.585; 26 0.122 -1.5 5.78501
   18. 34 0.122 0 12.385; 35 4.32201 0 12.385; 36 0.122 3.49961 12.385
   19. 37 4.32201 3.49961 12.385; 38 0.122 -1.5 12.385; 39 4.32201 -1.5 12.385
   20. 79 4.322 0 5.78501; 80 4.322 3.49961 5.78501; 81 4.322 -1.5 5.78501
   21. 82 0.122 3.49961 7.78501; 83 4.322 3.49961 7.78501; 84 0.122 3.49961 10.785
   22. 85 4.32201 3.49961 10.785
   23. MEMBER INCIDENCES
   24. 1 4 5; 3 1 4; 4 2 5; 8 7 10; 11 4 10; 25 1 2; 30 7 1; 31 1 23; 32 2 24
   25. 34 7 26; 488 7 34; 490 10 82; 497 36 37; 498 34 36; 499 35 37; 500 35 34
   26. 501 34 38; 502 35 39; 583 79 80; 584 79 2; 585 79 81; 586 79 35; 587 5 80
   27. 588 80 83; 589 7 79; 590 10 80; 591 82 84; 592 83 85; 593 82 83; 594 84 36
   28. 595 85 37; 596 84 85
   29. START GROUP DEFINITION
   30. MEMBER
   31. _COLUMN_C1 3 4 8 31 32 34 498 499 501 502 583 585
   32. _RB_1 11 490 587 588 591 592 594 595
   33. _RB_2 590
   34. _GB_1 25 30 488 500 584 586 589
   35. _RB-3 593 596
   36. _RB_4 1 497
   37. END GROUP DEFINITION
   38. DEFINE MATERIAL START
```



```
   39. ISOTROPIC CONCRETE
   40. E 2.17184E+007
   41. POISSON 0.17
   42. DENSITY 23.6158
   43. ALPHA 5E-006
   44. DAMP 0.05
   45. TYPE CONCRETE
   46. STRENGTH FCU 27578.9
   47. END DEFINE MATERIAL
   48. CONSTANTS
   49. MATERIAL CONCRETE ALL
   50. MEMBER PROPERTY AMERICAN
   51. 3 4 8 31 32 34 498 499 501 502 583 585 PRIS YD 0.4 ZD 0.4
   52. 1 497 PRIS YD 0.5 ZD 0.25
   53. MEMBER PROPERTY AMERICAN
   54. 25 30 488 500 584 586 589 PRIS YD 0.5 ZD 0.25
   55. MEMBER PROPERTY AMERICAN
   56. 11 490 587 588 591 592 594 595 PRIS YD 0.5 ZD 0.25
   57. MEMBER PROPERTY AMERICAN
   58. 590 593 596 PRIS YD 0.45 ZD 0.25
   59. SUPPORTS
   60. 23 24 26 38 39 81 FIXED
   61. ******************************************************************************
   62. LOAD 1 LOADTYPE DEAD  TITLE DL
   63. SELFWEIGHT Y -1
   64. MEMBER LOAD
   65. 25 30 488 500 584 586 589 UNI GY -12
   66. 11 490 587 588 591 592 594 595 UNI GY -13.45
   67. 11 490 591 594 UMOM GZ 1
   68. 587 588 592 595 UMOM GZ -1.
   69. 1 UMOM GX -1.
   70. 497 UMOM GX 1
   71. ******************************************************************************
   72. LOAD 2 LOADTYPE LIVE  TITLE LL
   73. MEMBER LOAD
   74. * LOAD DUE TO MONORAIL
   75. 593 596 CON GY -10 2.1
   76. 490 CON GY -10 1
   77. ******************************************************************************
   78. LOAD 3 LOADTYPE ROOF LIVE  TITLE LR
   79. MEMBER LOAD
   80. 11 490 587 588 591 592 594 595 UNI GY -1.05
   81. 11 490 591 594 UMOM GZ 0.1
   82. 587 588 592 595 UMOM GZ -0.1
   83. 1 UMOM GX -0.1
   84. 497 UMOM GX 0.1
   85. ******************************************************************************
   86. LOAD 4 LOADTYPE WIND  TITLE WIND ALONG X DIR. ( CPI = + 0.18) WLX1
   87. MEMBER LOAD
   88. 3 UNI GX 0.9
   89. 8 UNI GX 2.74
   90. 498 UNI GX 1.85
   91. 4 UNI GX 1.28
   92. 583 UNI GX 3.92
   93. 499 UNI GX 2.64
   94. 498 499 UNI GZ 1.74
```

```
 95. 3 4 UNI GZ -1.74
 96. ********************************************************************************
 97. LOAD 5 LOADTYPE WIND  TITLE WIND ALONG X DIR. ( CPI = - 0.18) WLX2
 98. MEMBER LOAD
 99. 3 UNI GX 1.65
100. 8 UNI GX 5.05
101. 498 UNI GX 3.4
102. 4 UNI GX 0.53
103. 583 UNI GX 1.62
104. 499 UNI GX 1.09
105. 498 499 UNI GZ 0.74
106. 3 4 UNI GZ -0.74
107. ********************************************************************************
108. LOAD 6 LOADTYPE WIND  TITLE WIND ALONG Z DIR. ( CPI = + 0.18) WLZ1
109. MEMBER LOAD
110. 3 UNI GX -1.33
111. 8 UNI GX -4.07
112. 498 UNI GX -2.74
113. 4 UNI GX 1.33
114. 583 UNI GX 4.07
115. 499 UNI GX 2.74
116. 498 499 UNI GZ 1.68
117. 3 4 UNI GZ 1.18
118. ********************************************************************************
119. LOAD 7 LOADTYPE WIND  TITLE WIND ALONG Z DIR. ( CPI = - 0.18) WLZ2
120. MEMBER LOAD
121. 3 UNI GX -0.56
122. 8 UNI GX -1.72
123. 498 UNI GX -1.16
124. 4 UNI GX 0.56
125. 583 UNI GX 1.72
126. 499 UNI GX 1.16
127. 498 499 UNI GZ 0.69
128. 3 4 UNI GZ 2.16
129. ********************************************************************************
130. LOAD 8 LOADTYPE TEMPERATURE  TITLE TEMPERATURE LOAD ( TL )
131. TEMPERATURE LOAD
132. 3 4 8 498 499 583 TEMP 25
133. 1 11 490 497 587 588 590 TO 596 TEMP 0 25
134. 3 4 8 498 499 583 TEMP 0 0 25
135. ********************************************************************************
136. * FACTORED LOAD COMBINATIONS
137. LOAD COMB 101 1.4DL
138. 1 1.4
139. LOAD COMB 102 1.4DL+1.7LL+0.5LR
140. 1 1.4 2 1.7 3 0.5
141. LOAD COMB 103 1.2DL+1.6LR+0.8WLX1
142. 1 1.2 3 1.6 4 0.8
143. LOAD COMB 104 1.2DL+1.6LR+0.8WLX2
144. 1 1.2 3 1.6 5 0.8
145. LOAD COMB 105 1.2DL+1.6LR+0.8WLZ1
146. 1 1.2 3 1.6 6 0.8
147. LOAD COMB 106 1.2DL+1.6LR+0.8WLZ2
148. 1 1.2 3 1.6 7 0.8
149. LOAD COMB 107 1.2DL+1.6LR+1.0LL
150. 1 1.2 2 1.0 3 1.6
```



```
151. LOAD COMB 108 1.2DL+1.0LL+1.6WLX1+0.5LR
152. 1 1.2 2 1.0 3 0.5 4 1.6
153. LOAD COMB 109 1.2DL+1.0LL+1.6WLX1+0.5LR
154. 1 1.2 2 1.0 3 0.5 5 1.6
155. LOAD COMB 110 1.2DL+1.0LL+1.6WLX1+0.5LR
156. 1 1.2 2 1.0 3 0.5 6 1.6
157. LOAD COMB 111 1.2DL+1.0LL+1.6WLX1+0.5LR
158. 1 1.2 2 1.0 3 0.5 7 1.6
159. LOAD COMB 112 1.2DL+1.0LL
160. 1 1.2 2 1.0
161. LOAD COMB 113 0.9DL+1.6WLX1
162. 1 0.9 4 1.6
163. LOAD COMB 114 0.9DL+1.6WLX2
164. 1 0.9 5 1.6
165. LOAD COMB 115 0.9DL+1.6WLZ1
166. 1 0.9 6 1.6
167. LOAD COMB 116 0.9DL+1.6WLZ2
168. 1 0.9 7 1.6
169. LOAD COMB 117 1.4DL+ 1.4TL+1.7LL+0.5LR
170. 1 1.4 2 1.7 3 0.5 8 1.4
171. ********************************************************************************
172. * UNFACTORED LOAD COMBINATIONS FOR BASE PRESSURE
173. LOAD COMB 201 1.0DL+1.0LL+1.0TL
174. 1 1.0 2 1.0 8 1.0
175. LOAD COMB 202 1.0DL+1.0LR
176. 1 1.0 3 1.0
177. LOAD COMB 203 1.0DL+0.75LL+0.75LR+0.75TL
178. 1 1.0 2 0.75 3 0.75 8 0.75
179. LOAD COMB 204 1.0DL+1.0WLX1
180. 1 1.0 4 1.0
181. LOAD COMB 205 1.0DL+1.0WLX2
182. 1 1.0 5 1.0
183. LOAD COMB 206 1.0DL+1.0WLZ1
184. 1 1.0 6 1.0
185. LOAD COMB 207 1.0DL+1.0WLZ2
186. 1 1.0 7 1.0
187. LOAD COMB 208 1.0DL+0.75LL+0.75LR+0.75WLX1
188. 1 1.0 2 0.75 3 0.75 4 0.75
189. LOAD COMB 209 1.0DL+0.75LL+0.75LR+0.75WLX2
190. 1 1.0 2 0.75 3 0.75 5 0.75
191. LOAD COMB 210 1.0DL+0.75LL+0.75LR+0.75WLZ1
192. 1 1.0 2 0.75 3 0.75 6 0.75
193. LOAD COMB 211 1.0DL+0.75LL+0.75LR+0.75WLZ2
194. 1 1.0 2 0.75 3 0.75 7 0.75
195. LOAD COMB 212 0.6DL+1.0WLX1
196. 1 0.6 4 1.0
197. LOAD COMB 213 0.6DL+1.0WLX2
198. 1 0.6 5 1.0
199. LOAD COMB 214 0.6DL+1.0WLZ1
200. 1 0.6 6 1.0
201. LOAD COMB 215 0.6DL+1.0WLZ2
202. 1 0.6 7 1.0
203. ********************************************************************************
204. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS 22 NUMBER OF MEMBERS 32
NUMBER OF PLATES 0 NUMBER OF SOLIDS 0
NUMBER OF SURFACES 0 NUMBER OF SUPPORTS 6

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 14/ 6/ 30 DOF
TOTAL PRIMARY LOAD CASES = 8, TOTAL DEGREES OF FREEDOM = 96
TOTAL LOAD COMBINATION CASES = 32 SO FAR.
SIZE OF STIFFNESS MATRIX = 3 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 12.1/ 187940.2 MB

```
205. LOAD LIST 101 TO 117
206. START CONCRETE DESIGN
207. CODE ACI
208. TRACK 2 ALL
209. FC 35000 ALL
210. CLB 0.04 ALL
211. CLS 0.04 ALL
212. CLT 0.04 ALL
213. MAXMAIN 20 MEMB 3 4 8 31 32 34 498 499 501 502 583 585
214. MINMAIN 20 MEMB 3 4 8 31 32 34 498 499 501 502 583 585
215. MAXMAIN 20 MEMB 1 11 490 497 587 588 590 TO 596
216. MINMAIN 20 MEMB 1 11 490 497 587 588 590 TO 596
217. MAXMAIN 16 MEMB 11 490 587 588 591 592 594 595
218. MINMAIN 16 MEMB 11 490 587 588 591 592 594 595
219. MAXMAIN 16 MEMB 25 30 488 500 584 586 589
220. MINMAIN 16 MEMB 25 30 488 500 584 586 589
221. MINSEC 10 ALL
222. DESIGN BEAM 1 11 25 30 488 490 497 500 584 586 TO 596
```


BEAM NO. 1 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	0.	4200.	YES	YES
	CRITICAL POS MOMENT= 15.40 KN-MET AT 2100.MM, LOAD 117					
	REQD STEEL= 126.MM2, RHO=0.0011, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS					
	REQD. DEVELOPMENT LENGTH = 505. MMS					
	Cracked Moment of Inertia Iz at above location = 59944.0 cm^4					
2	440.	2 - 20MM	0.	1357.	YES	NO
	CRITICAL NEG MOMENT= 6.40 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 52.MM2, RHO=0.0005, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS					
	REQD. DEVELOPMENT LENGTH = 505. MMS					
	Cracked Moment of Inertia Iz at above location = 59944.0 cm^4					
3	440.	2 - 20MM	1618.	4200.	NO	YES
	CRITICAL NEG MOMENT= 13.09 KN-MET AT 4200.MM, LOAD 109					
	REQD STEEL= 107.MM2, RHO=0.0010, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS					
	REQD. DEVELOPMENT LENGTH = 505. MMS					
	Cracked Moment of Inertia Iz at above location = 59944.0 cm^4					

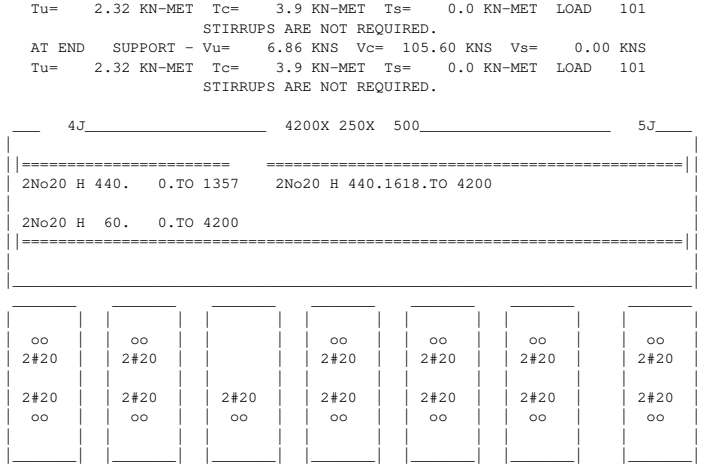
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	38./	40.	6./	6.	117/	102
350.	56./	22.	9./	4.	117/	102
700.	70./	8.	11./	1.	117/	102
1050.	81./	0.	13./	0.	117/	0
1400.	89./	0.	14./	0.	117/	0
1750.	94./	0.	15./	0.	117/	0
2100.	96./	0.	15./	0.	117/	0
2450.	94./	0.	15./	0.	117/	114

2800.	89./	11.	14./	2.	117/ 114
3150.	82./	23.	13./	4.	117/ 114
3500.	71./	39.	11./	6.	117/ 109
3850.	57./	59.	9./	9.	117/ 109
4200.	40./	81.	6./	13.	117/ 109

B E A M N O. 1 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.86 KNS Vc= 105.60 KNS Vs= 0.00 KNS
Tu= 2.32 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.86 KNS Vc= 105.60 KNS Vs= 0.00 KNS
Tu= 2.32 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 11 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	324.	2209.	NO	NO

CRITICAL POS MOMENT= 9.28 KN-MET AT 1600.MM, LOAD 117
REQD STEEL= 75.MM2, RHO=0.0007, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2 442. 2 - 16MM 0. 3200. YES YES

CRITICAL NEG MOMENT= 60.80 KN-MET AT 3200.MM, LOAD 102
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

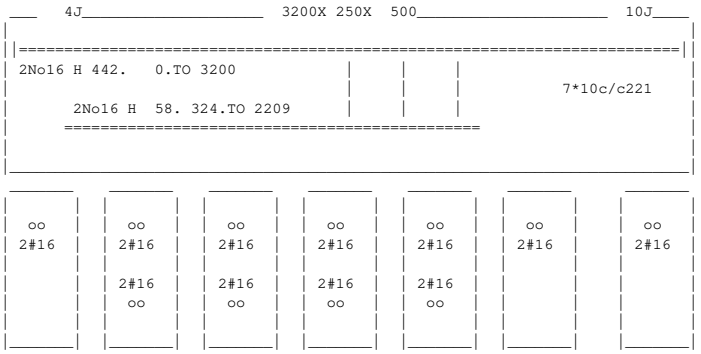
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 104.	0./ 17.	0/ 117
267.	0./ 51.	0./ 8.	0/ 117
533.	8./ 11.	1./ 2.	115/ 102
800.	23./ 1.	4./ 0.	117/ 102
1067.	45./ 2.	7./ 0.	117/ 102
1333.	56./ 12.	9./ 2.	117/ 102
1600.	57./ 33.	9./ 5.	117/ 102
1867.	48./ 65.	8./ 10.	117/ 102
2133.	28./ 107.	5./ 17.	117/ 102
2400.	0./ 159.	0./ 26.	0/ 102
2667.	0./ 223.	0./ 36.	0/ 102
2933.	0./ 298.	0./ 47.	0/ 102
3200.	0./ 384.	0./ 61.	0/ 102

B E A M N O. 11 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.58 KNS Vc= 107.61 KNS Vs= 0.00 KNS
Tu= 1.71 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 43.10 KNS Vc= 103.77 KNS Vs= 0.00 KNS
Tu= 1.72 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1160. MM



BEAM NO. 25 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	3.	3672.	NO	NO

CRITICAL POS MOMENT= 21.00 KN-MET AT 2100.MM, LOAD 117
REQD STEEL= 171.MM2, RHO=0.0015, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1226.	YES	NO

CRITICAL NEG MOMENT= 26.27 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 215.MM2, RHO=0.0019, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	2449.	4200.	NO	YES

CRITICAL NEG MOMENT= 33.84 KN-MET AT 4200.MM, LOAD 108
REQD STEEL= 277.MM2, RHO=0.0025, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

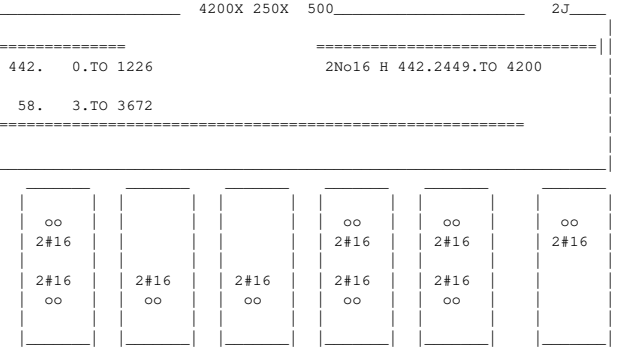
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	0./	164.	0./	26.	0/	102
350.	12./	75.	2./	12.	114/	102
700.	46./	6.	7./	1.	114/	110
1050.	81./	0.	13./	0.	109/	0
1400.	104./	0.	17./	0.	109/	0
1750.	122./	0.	20./	0.	117/	0
2100.	131./	0.	21./	0.	117/	0
2450.	123./	0.	20./	0.	117/	0
2800.	99./	0.	16./	0.	117/	0
3150.	59./	2.	10./	0.	117/	113
3500.	3./	50.	1./	8.	117/	108
3850.	0./	123.	0./	20.	0/	108
4200.	0./	211.	0./	34.	0/	108

B E A M N O. 25 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.75 KNS Vc= 109.95 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 34.75 KNS Vc= 109.95 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 30 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	847.	3020.	NO	NO

CRITICAL POS MOMENT= 10.61 KN-MET AT 2133.MM, LOAD 117
REQD STEEL= 86.MM2, RHO=0.0008, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1592.	YES	NO

CRITICAL NEG MOMENT= 33.93 KN-MET AT 0.MM, LOAD 117
REQD STEEL= 278.MM2, RHO=0.0025, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	2008.	3200.	NO	YES

CRITICAL NEG MOMENT= 16.48 KN-MET AT 3200.MM, LOAD 102
REQD STEEL= 134.MM2, RHO=0.0012, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

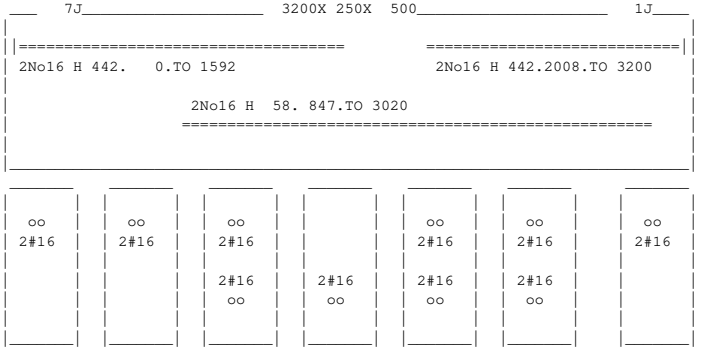
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 212.	0./ 34.	0/ 117
267.	0./ 144.	0./ 23.	0/ 117
533.	0./ 86.	0./ 14.	0/ 117
800.	0./ 37.	0./ 6.	0/ 117
1067.	20./ 2.	3./ 0.	102/ 115
1333.	37./ 0.	6./ 0.	102/ 0
1600.	52./ 0.	8./ 0.	117/ 0
1867.	63./ 0.	10./ 0.	117/ 0
2133.	66./ 0.	11./ 0.	117/ 0
2400.	59./ 0.	9./ 0.	117/ 0
2667.	42./ 18.	7./ 3.	117/ 108
2933.	17./ 54.	3./ 9.	117/ 102
3200.	0./ 102.	0./ 16.	0/ 102

B E A M N O. 30 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 27.07 KNS Vc= 104.83 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.50 KNS Vc= 106.67 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 488 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6600. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	1028.	5297.	NO	NO

CRITICAL POS MOMENT= 42.05 KN-MET AT 3300.MM, LOAD 101
REQD STEEL= 345.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	3 - 16MM	0.	1626.	YES	NO

CRITICAL NEG MOMENT= 77.03 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 483.MM2, RHO=0.0044, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

3	442.	3 - 16MM	4699.	6600.	NO	YES

CRITICAL NEG MOMENT= 74.31 KN-MET AT 6600.MM, LOAD 117
REQD STEEL= 465.MM2, RHO=0.0042, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

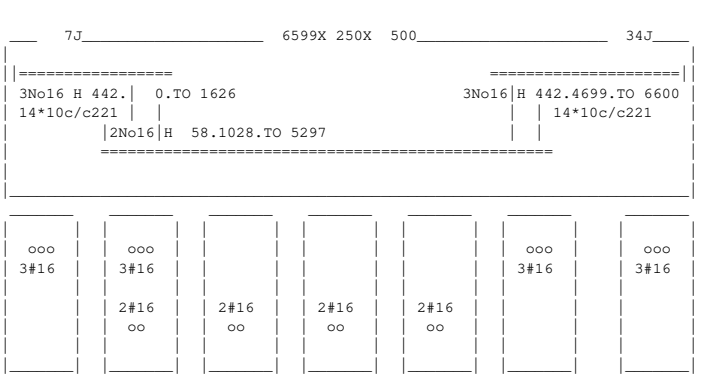
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	0./	490.	0./	77.	0/	102
550.	0./	260.	0./	41.	0/	102
1100.	0./	75.	0./	12.	0/	102
1650.	82./	0.	13./	0.	117/	0
2200.	180./	0.	29./	0.	117/	0
2750.	239./	0.	38./	0.	117/	0
3300.	264./	0.	42./	0.	101/	0
3850.	249./	0.	40./	0.	101/	0
4400.	194./	0.	31./	0.	101/	0
4950.	99./	0.	16./	0.	101/	0
5500.	0./	65.	0./	11.	0/	117
6050.	0./	246.	0./	39.	0/	117
6600.	0./	472.	0./	74.	0/	117

B E A M N O. 488 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 61.34 KNS Vc= 106.49 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2860. MM

AT END SUPPORT - Vu= 58.39 KNS Vc= 107.01 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2860. MM



BEAM NO. 490 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	642.	2000.	NO	YES

CRITICAL POS MOMENT= 65.74 KN-MET AT 2000.MM, LOAD 117
REQD STEEL= 410.MM2, RHO=0.0037, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

2	442.	4 - 16MM	0.	1860.	YES	NO

CRITICAL NEG MOMENT= 115.61 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 737.MM2, RHO=0.0067, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 45. MMS
REQD. DEVELOPMENT LENGTH = 724. MMS

Cracked Moment of Inertia Iz at above location = 73952.4 cm^4

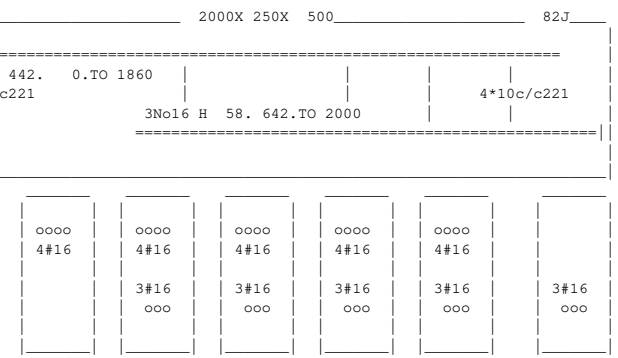
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 748.	0./ 116.	0/ 102
167.	0./ 623.	0./ 97.	0/ 102
333.	0./ 504.	0./ 79.	0/ 102
500.	0./ 391.	0./ 62.	0/ 102
667.	0./ 283.	0./ 45.	0/ 102
833.	0./ 182.	0./ 29.	0/ 102
1000.	21./ 109.	3./ 17.	117/ 101
1167.	96./ 43.	15./ 7.	117/ 101
1333.	167./ 0.	27./ 0.	117/ 0
1500.	235./ 0.	38./ 0.	117/ 0
1667.	299./ 0.	48./ 0.	117/ 0
1833.	360./ 0.	57./ 0.	117/ 0
2000.	416./ 0.	66./ 0.	117/ 0

B E A M N O. 490 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.80 KNS Vc= 107.79 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 560. MM

AT END SUPPORT - Vu= 52.09 KNS Vc= 119.55 KNS Vs= 0.00 KNS
Tu= 1.54 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 560. MM



BEAM NO. 497 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	0.	4200.	YES	YES

CRITICAL POS MOMENT= 15.84 KN-MET AT 2100.MM, LOAD 117
REQD STEEL= 129.MM2, RHO=0.0012, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS
REQD. DEVELOPMENT LENGTH = 505. MMS

Cracked Moment of Inertia Iz at above location = 59944.0 cm^4

2	440.	2 - 20MM	0.	1357.	YES	NO

CRITICAL NEG MOMENT= 5.64 KN-MET AT 0.MM, LOAD 101
REQD STEEL= 46.MM2, RHO=0.0004, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS
REQD. DEVELOPMENT LENGTH = 505. MMS

Cracked Moment of Inertia Iz at above location = 59944.0 cm^4

3	440.	2 - 20MM	1618.	4200.	NO	YES

CRITICAL NEG MOMENT= 13.58 KN-MET AT 4200.MM, LOAD 109
REQD STEEL= 111.MM2, RHO=0.0010, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS
REQD. DEVELOPMENT LENGTH = 505. MMS

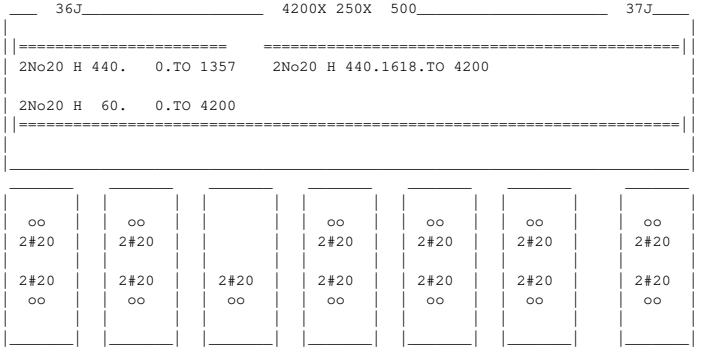
Cracked Moment of Inertia Iz at above location = 59944.0 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	41./ 35.	7./ 6.	117/ 101
350.	58./ 18.	9./ 3.	117/ 101
700.	72./ 4.	12./ 1.	117/ 101
1050.	84./ 0.	13./ 0.	117/ 0
1400.	92./ 0.	15./ 0.	117/ 0
1750.	97./ 0.	16./ 0.	117/ 0
2100.	98./ 0.	16./ 0.	117/ 0
2450.	97./ 1.	16./ 0.	117/ 114
2800.	92./ 13.	15./ 2.	117/ 114
3150.	84./ 27.	14./ 4.	117/ 114
3500.	74./ 43.	12./ 7.	117/ 114
3850.	60./ 61.	10./ 10.	117/ 114
4200.	42./ 84.	7./ 14.	117/ 109

B E A M N O. 497 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.86 KNS Vc= 107.23 KNS Vs= 0.00 KNS
Tu= 2.32 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.86 KNS Vc= 107.23 KNS Vs= 0.00 KNS
Tu= 2.32 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 500 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	354.	4200.	NO	YES

CRITICAL POS MOMENT= 20.94 KN-MET AT 2100.MM, LOAD 117
REQD STEEL= 171.MM2, RHO=0.0015, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1576.	YES	NO

CRITICAL NEG MOMENT= 39.89 KN-MET AT 0.MM, LOAD 108
REQD STEEL= 327.MM2, RHO=0.0030, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	2799.	4200.	NO	YES

CRITICAL NEG MOMENT= 26.41 KN-MET AT 4200.MM, LOAD 102
REQD STEEL= 216.MM2, RHO=0.0020, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

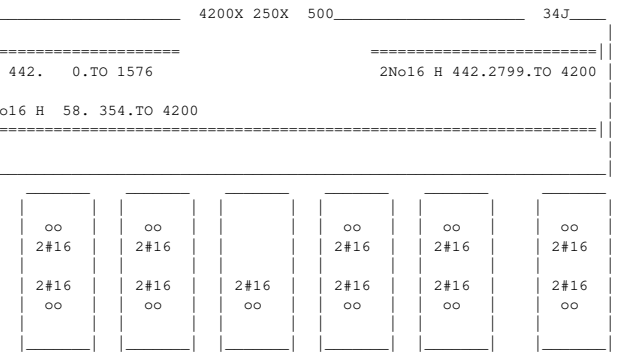
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 250.	0./ 40.	0/ 108
350.	0./ 155.	0./ 25.	0/ 108
700.	3./ 76.	0./ 12.	117/ 108
1050.	58./ 21.	9./ 3.	117/ 113
1400.	98./ 0.	16./ 0.	117/ 0
1750.	122./ 0.	20./ 0.	117/ 0
2100.	130./ 0.	21./ 0.	117/ 0
2450.	122./ 0.	20./ 0.	117/ 0
2800.	116./ 0.	19./ 0.	109/ 0
3150.	100./ 0.	16./ 0.	109/ 0
3500.	71./ 10.	11./ 2.	114/ 110
3850.	43./ 76.	7./ 12.	114/ 102
4200.	4./ 165.	1./ 26.	114/ 102

B E A M N O. 500 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.75 KNS Vc= 109.69 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 34.75 KNS Vc= 109.69 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 584 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	848.	3019.	NO	NO

CRITICAL POS MOMENT= 10.49 KN-MET AT 2133.MM, LOAD 117
REQD STEEL= 85.MM2, RHO=0.0008, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1592.	YES	NO

CRITICAL NEG MOMENT= 34.47 KN-MET AT 0.MM, LOAD 117
REQD STEEL= 282.MM2, RHO=0.0026, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	2008.	3200.	NO	YES

CRITICAL NEG MOMENT= 16.38 KN-MET AT 3200.MM, LOAD 102
REQD STEEL= 133.MM2, RHO=0.0012, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

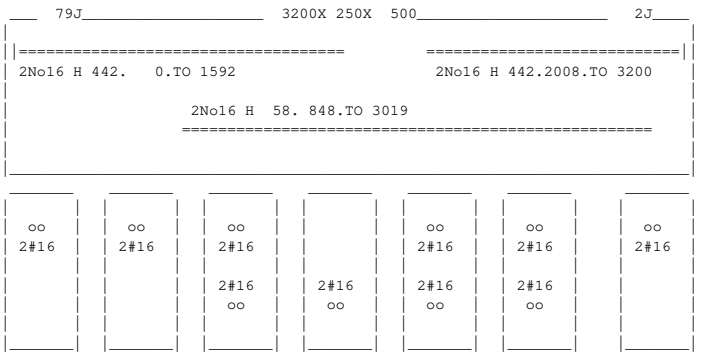
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 215.	0./ 34.	0/ 117
267.	0./ 147.	0./ 24.	0/ 117
533.	0./ 89.	0./ 14.	0/ 117
800.	0./ 40.	0./ 6.	0/ 117
1067.	18./ 2.	3./ 0.	102/ 115
1333.	35./ 0.	6./ 0.	102/ 0
1600.	51./ 0.	8./ 0.	117/ 0
1867.	62./ 0.	10./ 0.	117/ 0
2133.	65./ 0.	10./ 0.	117/ 0
2400.	58./ 0.	9./ 0.	117/ 0
2667.	42./ 21.	7./ 3.	117/ 108
2933.	17./ 55.	3./ 9.	117/ 108
3200.	0./ 102.	0./ 16.	0/ 102

B E A M N O. 584 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 27.07 KNS Vc= 104.82 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 21.50 KNS Vc= 106.63 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 586 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 6600. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	1028.	5297.	NO	NO

CRITICAL POS MOMENT= 42.05 KN-MET AT 3300.MM, LOAD 101
REQD STEEL= 345.MM2, RHO=0.0031, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	3 - 16MM	0.	1626.	YES	NO

CRITICAL NEG MOMENT= 76.98 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 482.MM2, RHO=0.0044, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

3	442.	3 - 16MM	4699.	6600.	NO	YES

CRITICAL NEG MOMENT= 74.10 KN-MET AT 6600.MM, LOAD 117
REQD STEEL= 464.MM2, RHO=0.0042, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

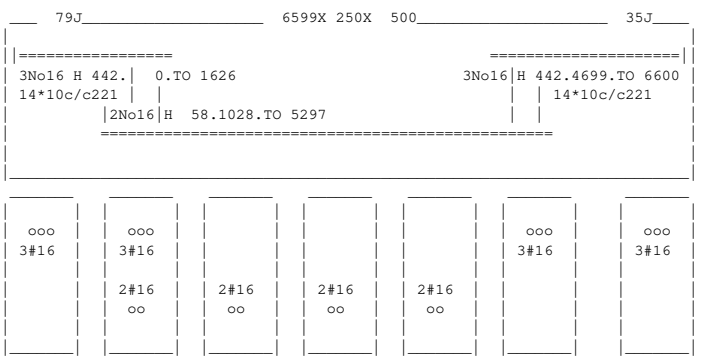
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	0./	490.	0./	77.	0/	102
550.	0./	259.	0./	41.	0/	102
1100.	0./	75.	0./	12.	0/	102
1650.	83./	0.	13./	0.	117/	0
2200.	181./	0.	29./	0.	117/	0
2750.	239./	0.	38./	0.	117/	0
3300.	264./	0.	42./	0.	101/	0
3850.	249./	0.	40./	0.	101/	0
4400.	194./	0.	31./	0.	101/	0
4950.	99./	0.	16./	0.	101/	0
5500.	0./	64.	0./	10.	0/	117
6050.	0./	245.	0./	39.	0/	117
6600.	0./	471.	0./	74.	0/	117

B E A M N O. 586 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 61.34 KNS Vc= 106.49 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2860. MM

AT END SUPPORT - Vu= 58.39 KNS Vc= 107.01 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 3.8 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 2860. MM



BEAM NO. 587 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	310.	2490.	NO	NO

CRITICAL POS MOMENT= 11.06 KN-MET AT 1600.MM, LOAD 117
REQD STEEL= 90.MM2, RHO=0.0008, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1059.	YES	NO

CRITICAL NEG MOMENT= 17.79 KN-MET AT 0.MM, LOAD 117
REQD STEEL= 145.MM2, RHO=0.0013, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	674.	3200.	NO	YES

CRITICAL NEG MOMENT= 56.19 KN-MET AT 3200.MM, LOAD 102
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

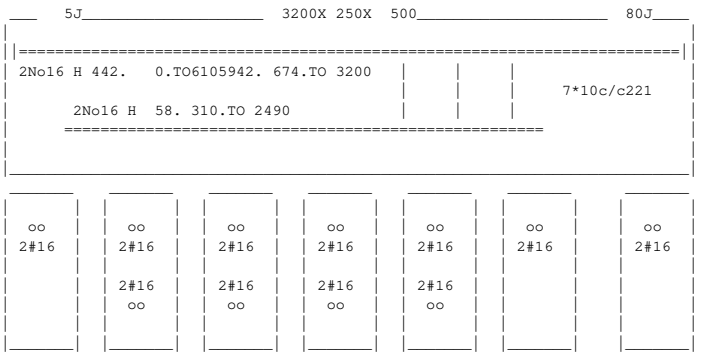
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	0./	110.	0./	18.	0/	117
267.	0./	54.	0./	9.	0/	117
533.	8./	12.	1./	2.	115/	102
800.	26./	0.	4./	0.	117/	0
1067.	50./	0.	8./	0.	117/	0
1333.	65./	4.	10./	1.	117/	102
1600.	68./	22.	11./	4.	117/	102
1867.	62./	51.	10./	8.	117/	102
2133.	45./	90.	7./	14.	117/	102
2400.	18./	139.	3./	22.	117/	102
2667.	0./	200.	0./	32.	0/	102
2933.	0./	271.	0./	43.	0/	102
3200.	0./	354.	0./	56.	0/	102

B E A M N O. 587 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 14.58 KNS Vc= 108.46 KNS Vs= 0.00 KNS
Tu= 1.71 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 41.32 KNS Vc= 103.95 KNS Vs= 0.00 KNS
Tu= 1.79 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 102
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 1160. MM



BEAM NO. 588 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	688.	2000.	NO	YES

CRITICAL POS MOMENT= 61.61 KN-MET AT 2000.MM, LOAD 117
REQD STEEL= 394.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	4 - 16MM	0.	1860.	YES	NO

CRITICAL NEG MOMENT= 106.55 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 676.MM2, RHO=0.0061, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 45. MMS
REQD. DEVELOPMENT LENGTH = 724. MMS

Cracked Moment of Inertia Iz at above location = 73952.4 cm^4

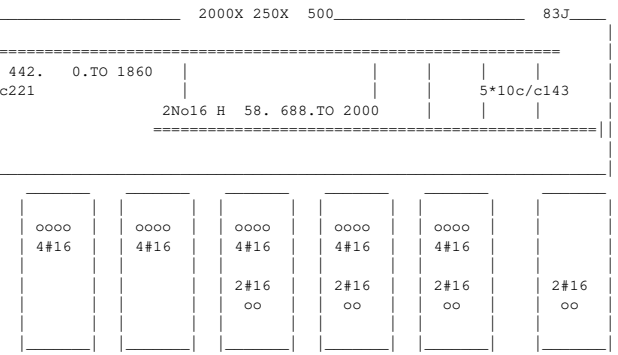
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 687.	0./ 107.	0/ 102
167.	0./ 579.	0./ 90.	0/ 102
333.	0./ 477.	0./ 75.	0/ 102
500.	0./ 381.	0./ 60.	0/ 102
667.	0./ 290.	0./ 46.	0/ 102
833.	0./ 204.	0./ 33.	0/ 102
1000.	0./ 123.	0./ 20.	0/ 102
1167.	60./ 47.	10./ 8.	117/ 102
1333.	133./ 0.	21./ 0.	117/ 0
1500.	203./ 0.	32./ 0.	117/ 0
1667.	269./ 0.	43./ 0.	117/ 0
1833.	331./ 0.	53./ 0.	117/ 0
2000.	389./ 0.	62./ 0.	117/ 0

B E A M N O. 588 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 77.80 KNS Vc= 107.60 KNS Vs= 0.00 KNS
Tu= 0.03 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 560. MM

AT END SUPPORT - Vu= 61.81 KNS Vc= 117.95 KNS Vs= 0.00 KNS
Tu= 4.29 KN-MET Tc= 4.1 KN-MET Ts= 5.7 KN-MET LOAD 102
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 560. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.41 SQ.CM.



BEAM NO. 589 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	3672.	YES	NO

CRITICAL POS MOMENT= 21.01 KN-MET AT 2100.MM, LOAD 117
REQD STEEL= 171.MM2, RHO=0.0015, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	2 - 16MM	0.	1226.	YES	NO

CRITICAL NEG MOMENT= 26.21 KN-MET AT 0.MM, LOAD 102
REQD STEEL= 214.MM2, RHO=0.0019, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

3	442.	2 - 16MM	2449.	4200.	NO	YES

CRITICAL NEG MOMENT= 42.58 KN-MET AT 4200.MM, LOAD 108
REQD STEEL= 350.MM2, RHO=0.0032, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

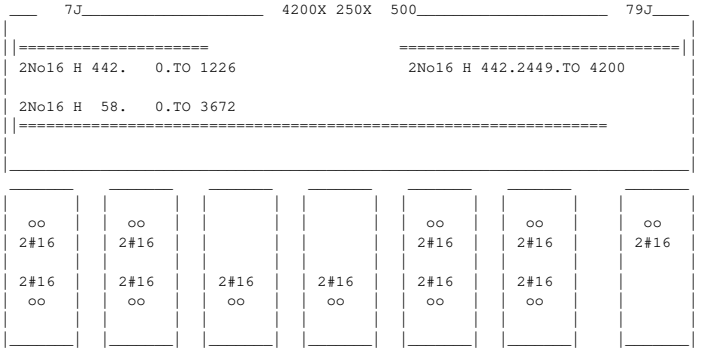
Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	23./	163.	4./	26.	114/	102
350.	59./	75.	9./	12.	114/	102
700.	84./	12.	14./	2.	114/	110
1050.	111./	0.	18./	0.	109/	0
1400.	124./	0.	20./	0.	109/	0
1750.	124./	0.	20./	0.	109/	0
2100.	131./	0.	21./	0.	117/	0
2450.	123./	0.	20./	0.	117/	0
2800.	98./	0.	16./	0.	117/	0
3150.	59./	29.	9./	5.	117/	113
3500.	3./	87.	0./	14.	117/	108
3850.	0./	169.	0./	27.	0/	108
4200.	0./	267.	0./	43.	0/	108

B E A M N O. 589 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 34.75 KNS Vc= 110.12 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 34.75 KNS Vc= 110.12 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.9 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 590 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	0.	4200.	YES	YES

CRITICAL POS MOMENT= 13.29 KN-MET AT 2100.MM, LOAD 117
REQD STEEL= 123.MM2, RHO=0.0013, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS
REQD. DEVELOPMENT LENGTH = 505. MMS

Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

2	390.	2 - 20MM	0.	1357.	YES	NO

CRITICAL NEG MOMENT= 5.97 KN-MET AT 0.MM, LOAD 101
REQD STEEL= 55.MM2, RHO=0.0006, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS
REQD. DEVELOPMENT LENGTH = 505. MMS

Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

3	390.	2 - 20MM	1618.	4200.	NO	YES

CRITICAL NEG MOMENT= 13.16 KN-MET AT 4200.MM, LOAD 109
REQD STEEL= 121.MM2, RHO=0.0012, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS
REQD. DEVELOPMENT LENGTH = 505. MMS

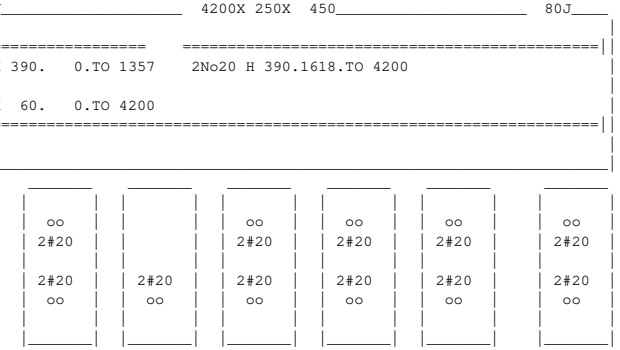
Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	35./ 42.	5./ 6.	117/ 101
350.	52./ 24.	7./ 3.	117/ 101
700.	67./ 10.	10./ 1.	117/ 101
1050.	78./ 0.	11./ 0.	117/ 0
1400.	86./ 0.	12./ 0.	117/ 0
1750.	91./ 0.	13./ 0.	117/ 0
2100.	93./ 0.	13./ 0.	117/ 0
2450.	92./ 6.	13./ 1.	117/ 114
2800.	87./ 18.	12./ 3.	117/ 114
3150.	79./ 32.	11./ 5.	117/ 114
3500.	68./ 48.	10./ 7.	117/ 114
3850.	54./ 68.	8./ 10.	117/ 109
4200.	36./ 92.	5./ 13.	117/ 109

B E A M N O. 590 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.36 KNS Vc= 92.98 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.4 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.36 KNS Vc= 92.98 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.4 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 591 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	0.	3000.	YES	YES

CRITICAL POS MOMENT= 90.91 KN-MET AT 1500.MM, LOAD 117
REQD STEEL= 573.MM2, RHO=0.0052, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

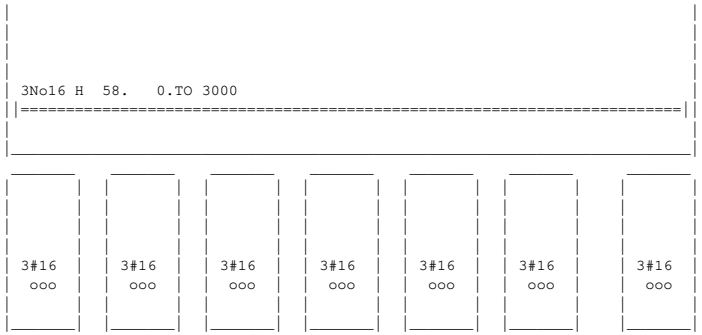
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	416./	0.	66./	0.	117/	0
250.	467./	0.	74./	0.	117/	0
500.	509./	0.	80./	0.	117/	0
750.	542./	0.	85./	0.	117/	0
1000.	565./	0.	88./	0.	117/	0
1250.	578./	0.	90./	0.	117/	0
1500.	582./	0.	91./	0.	117/	0
1750.	576./	0.	90./	0.	117/	0

2000.	560./	0.	88./	0.	117/	0
2250.	534./	0.	84./	0.	117/	0
2500.	499./	0.	78./	0.	117/	0
2750.	454./	0.	72./	0.	117/	0
3000.	401./	0.	63./	0.	117/	0

B E A M N O. 591 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 24.07 KNS Vc= 105.56 KNS Vs= 0.00 KNS
Tu= 1.41 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 24.61 KNS Vc= 105.65 KNS Vs= 0.00 KNS
Tu= 1.55 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 592 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 3000. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	3 - 16MM	0.	3000.	YES	YES

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|-------------------------------------------------------------------|
|   CRITICAL POS MOMENT=     89.17 KN-MET  AT  1500.MM, LOAD  117|
|   REQD STEEL=   562.MM2, RHO=0.0051, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS         |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                        |
|-------------------------------------------------------------------|
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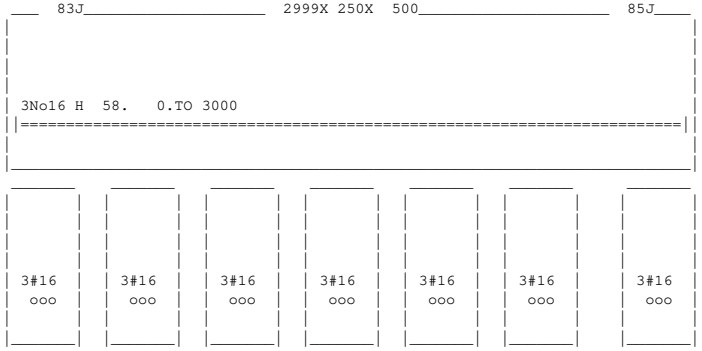
Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	390./	0.	62./	0.	117/	0
250.	444./	0.	70./	0.	117/	0
500.	488./	0.	77./	0.	117/	0
750.	523./	0.	82./	0.	117/	0
1000.	549./	0.	86./	0.	117/	0
1250.	564./	0.	88./	0.	117/	0
1500.	570./	0.	89./	0.	117/	0
1750.	567./	0.	89./	0.	117/	0
2000.	553./	0.	87./	0.	117/	0
2250.	530./	0.	83./	0.	117/	0
2500.	497./	0.	78./	0.	117/	0
2750.	455./	0.	72./	0.	117/	0
3000.	404./	0.	64./	0.	117/	0

B E A M N O. 592 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   24.07 KNS  Vc=  105.51 KNS  Vs=    0.00 KNS
Tu=    1.41 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   24.61 KNS  Vc=  105.60 KNS  Vs=    0.00 KNS
Tu=    1.55 KN-MET  Tc=    4.0 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                    STIRRUPS ARE NOT REQUIRED.
```

BEAM NO. 593 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	8.	4017.	NO	NO

```
|-------------------------------------------------------------------|
|   CRITICAL POS MOMENT=     22.41 KN-MET  AT  2100.MM, LOAD  117|
|   REQD STEEL=   208.MM2, RHO=0.0021, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   45./  130. MMS         |
|   REQD. DEVELOPMENT LENGTH =   505. MMS                        |
|-------------------------------------------------------------------|
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Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

2	390.	2 - 20MM	0.	1007.	YES	NO

```
|-------------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      6.34 KN-MET  AT     0.MM, LOAD  102|
|   REQD STEEL=    58.MM2, RHO=0.0006, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   45./  130. MMS         |
|   REQD. DEVELOPMENT LENGTH =   505. MMS                        |
|-------------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

3	390.	2 - 20MM	2668.	4200.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      7.24 KN-MET  AT  4200.MM, LOAD  102|
|   REQD STEEL=    67.MM2, RHO=0.0007, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   45./  130. MMS        |
|   REQD. DEVELOPMENT LENGTH =   505. MMS                      |
|-------------------------------------------------------------|
```

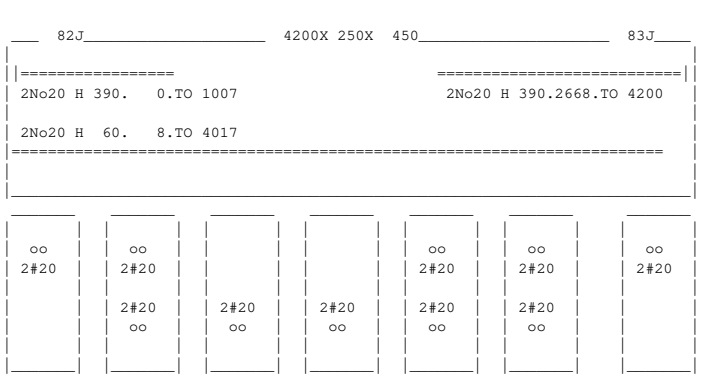
Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	0./	44.	0./	6.	0/	102
350.	15./	12.	2./	2.	117/	101
700.	50./	0.	7./	0.	117/	0
1050.	82./	0.	12./	0.	117/	0
1400.	110./	0.	16./	0.	117/	0
1750.	136./	0.	19./	0.	117/	0
2100.	158./	0.	22./	0.	117/	0
2450.	135./	0.	19./	0.	117/	0
2800.	108./	0.	15./	0.	117/	0
3150.	78./	0.	11./	0.	117/	0
3500.	46./	4.	7./	1.	117/	114
3850.	10./	17.	1./	2.	117/	109
4200.	0./	51.	0./	7.	0/	102

B E A M N O. 593 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    6.36 KNS  Vc=   97.16 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    3.3 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=    6.36 KNS  Vc=   97.16 KNS  Vs=    0.00 KNS
Tu=    0.00 KN-MET  Tc=    3.3 KN-MET  Ts=    0.0 KN-MET  LOAD   101
                  STIRRUPS ARE NOT REQUIRED.
```

BEAM NO. 594 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1600. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	1331.	YES	NO

```
|-------------------------------------------------------------|
|   CRITICAL POS MOMENT=     63.55 KN-MET  AT     0.MM, LOAD  117|
|   REQD STEEL=   396.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./  134. MMS        |
|   REQD. DEVELOPMENT LENGTH =   404. MMS                      |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

2	442.	3 - 16MM	74.	1600.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     72.91 KN-MET  AT  1600.MM, LOAD  102|
|   REQD STEEL=   456.MM2, RHO=0.0041, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   41./   67. MMS        |
|   REQD. DEVELOPMENT LENGTH =   483. MMS                      |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

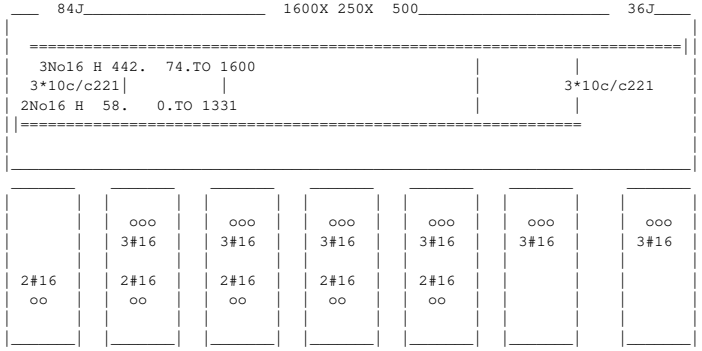
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	402./	0.	64./	0.	117/	0
133.	356./	0.	56./	0.	117/	0
267.	307./	0.	49./	0.	117/	0
400.	255./	0.	41./	0.	117/	0
533.	202./	0.	32./	0.	117/	0
667.	146./	6.	23./	1.	117/	116
800.	88./	52.	14./	8.	117/	111
933.	28./	109.	5./	18.	117/	102
1067.	0./	173.	0./	28.	0/	102
1200.	0./	241.	0./	38.	0/	102
1333.	0./	311.	0./	50.	0/	102
1467.	0./	386.	0./	61.	0/	102
1600.	0./	463.	0./	73.	0/	102

B E A M N O. 594 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 52.63 KNS Vc= 121.02 KNS Vs= 0.00 KNS
Tu= 1.15 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 360. MM

AT END SUPPORT - Vu= 69.17 KNS Vc= 111.69 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 360. MM



BEAM NO. 595 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 1600. MM FY - 414. FC - 35. MPA, SIZE - 250. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	58.	2 - 16MM	0.	1331.	YES	NO

CRITICAL POS MOMENT= 63.57 KN-MET AT 0.MM, LOAD 117
REQD STEEL= 396.MM2, RHO=0.0036, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 134. MMS
REQD. DEVELOPMENT LENGTH = 404. MMS

Cracked Moment of Inertia Iz at above location = 41662.2 cm^4

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
2	442.	3 - 16MM	74.	1600.	NO	YES

CRITICAL NEG MOMENT= 70.63 KN-MET AT 1600.MM, LOAD 102
REQD STEEL= 441.MM2, RHO=0.0040, RHOMX=0.0254 RHOMN=0.0036
MAX/MIN/ACTUAL BAR SPACING= 256./ 41./ 67. MMS
REQD. DEVELOPMENT LENGTH = 483. MMS

Cracked Moment of Inertia Iz at above location = 58557.4 cm^4

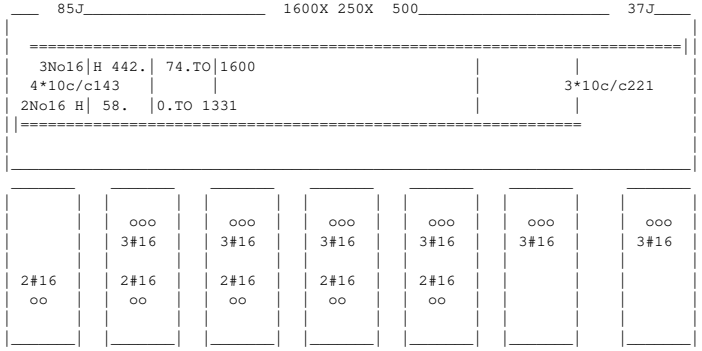
REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)		MOMENTS(+VE/-VE) (KNS-MET)		LOAD(+VE/-VE)	
0.	402./	0.	64./	0.	117/	0
133.	357./	0.	57./	0.	117/	0
267.	309./	0.	49./	0.	117/	0
400.	259./	0.	41./	0.	117/	0
533.	207./	0.	33./	0.	117/	0
667.	152./	6.	24./	1.	117/	116
800.	95./	48.	15./	8.	117/	111
933.	36./	101.	6./	16.	117/	102
1067.	0./	164.	0./	26.	0/	102
1200.	0./	230.	0./	37.	0/	102
1333.	0./	299.	0./	48.	0/	102
1467.	0./	372.	0./	59.	0/	102
1600.	0./	448.	0./	71.	0/	102

B E A M N O. 595 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 62.19 KNS Vc= 119.61 KNS Vs= 0.00 KNS
Tu= 4.18 KN-MET Tc= 4.1 KN-MET Ts= 5.6 KN-MET LOAD 102
STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 143. MM C/C FOR 360. MM
ADDITIONAL LONGITUDINAL STEEL REQD. FOR TORSIONAL RESISTANCE = 1.38 SQ.CM.

AT END SUPPORT - Vu= 69.17 KNS Vc= 111.69 KNS Vs= 0.00 KNS
Tu= 0.14 KN-MET Tc= 4.0 KN-MET Ts= 0.0 KN-MET LOAD 101
NO STIRRUPS ARE REQUIRED FOR TORSION.
REINFORCEMENT FOR SHEAR IS PER CL.11.5.5.1.
PROVIDE 10 MM 2-LEGGED STIRRUPS AT 221. MM C/C FOR 360. MM



BEAM NO. 596 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 4200. MM FY - 414. FC - 35. MPA, SIZE - 250. X 450. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	60.	2 - 20MM	6.	4019.	NO	NO
	CRITICAL POS MOMENT= 22.63 KN-MET AT 2100.MM, LOAD 117					
	REQD STEEL= 210.MM2, RHO=0.0022, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS					
	REQD. DEVELOPMENT LENGTH = 505. MMS					
	Cracked Moment of Inertia Iz at above location = 46054.1 cm^4					
2	390.	2 - 20MM	0.	1007.	YES	NO
	CRITICAL NEG MOMENT= 6.73 KN-MET AT 0.MM, LOAD 102					
	REQD STEEL= 62.MM2, RHO=0.0006, RHOMX=0.0254 RHOMN=0.0036					
	MAX/MIN/ACTUAL BAR SPACING= 256./ 45./ 130. MMS					
	REQD. DEVELOPMENT LENGTH = 505. MMS					
	Cracked Moment of Inertia Iz at above location = 46054.1 cm^4					
3	390.	2 - 20MM	2668.	4200.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      6.86 KN-MET  AT  4200.MM, LOAD  102|
|   REQD STEEL=    63.MM2, RHO=0.0006, RHOMX=0.0254 RHOMN=0.0036 |
|   MAX/MIN/ACTUAL BAR SPACING=   256./   45./  130. MMS        |
|   REQD. DEVELOPMENT LENGTH =   505. MMS                       |
|-------------------------------------------------------------|
```

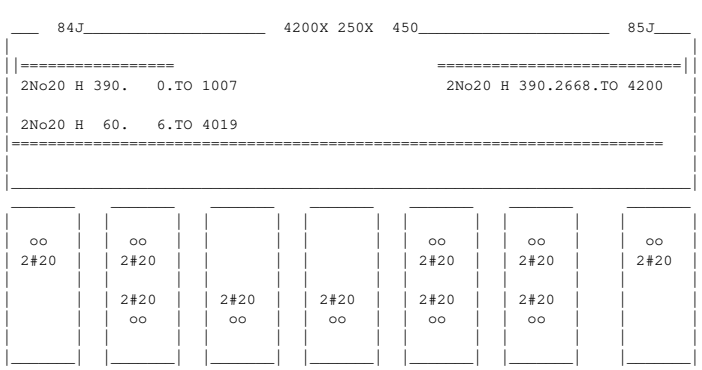
Cracked Moment of Inertia Iz at above location = 46054.1 cm^4

REQUIRED REINF. STEEL SUMMARY :

SECTION (MM)	REINF STEEL(+VE/-VE) (SQ. MM)	MOMENTS(+VE/-VE) (KNS-MET)	LOAD(+VE/-VE)
0.	0./ 47.	0./ 7.	0/ 102
350.	15./ 10.	2./ 1.	117/ 101
700.	50./ 0.	7./ 0.	117/ 0
1050.	82./ 0.	12./ 0.	117/ 0
1400.	111./ 0.	16./ 0.	117/ 0
1750.	137./ 0.	19./ 0.	117/ 0
2100.	159./ 0.	23./ 0.	117/ 0
2450.	137./ 0.	19./ 0.	117/ 0
2800.	111./ 0.	16./ 0.	117/ 0
3150.	81./ 0.	12./ 0.	117/ 0
3500.	49./ 3.	7./ 0.	117/ 114
3850.	14./ 15.	2./ 2.	117/ 109
4200.	0./ 48.	0./ 7.	0/ 102

B E A M N O. 596 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 6.36 KNS Vc= 98.55 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.3 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 6.36 KNS Vc= 98.55 KNS Vs= 0.00 KNS
Tu= 0.00 KN-MET Tc= 3.3 KN-MET Ts= 0.0 KN-MET LOAD 101
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

223. DESIGN COLUMN 3 4 8 31 32 34 498 499 501 502 583 585

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	51.32	6.13	0.03411

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	51.32	12.94	0.07207

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens	MOMENT				

COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	51.32	6.13	0.03411

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	51.32	12.94	0.07207

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens	MOMENT				

COLUMN NO. 8 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	222.33	4.88	0.00627

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	222.33	67.66	0.08695

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens MOMENT					

COLUMN NO. 31 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	206.89	10.67	0.03438

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	206.89	7.99	0.02576

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens MOMENT					

COLUMN NO. 32 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

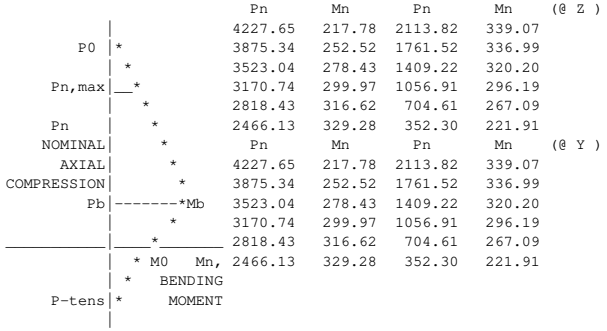
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	206.89	10.67	0.03438

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	206.89	7.99	0.02576



COLUMN NO. 34 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

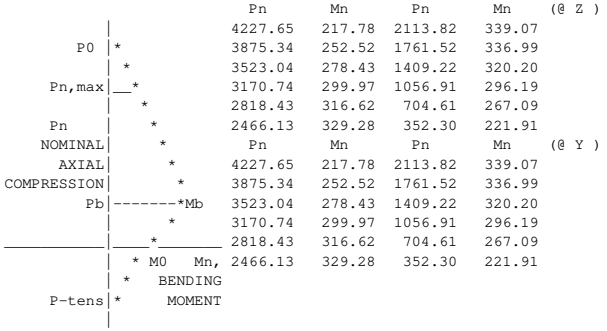
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	495.00	10.77	0.01450

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	495.00	17.54	0.02363



COLUMN NO. 498 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	135.30	7.94	0.01677

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	135.30	91.31	0.19283

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens	MOMENT				

COLUMN NO. 499 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	135.30	7.94	0.01677

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	135.30	91.31	0.19283

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens	MOMENT				

COLUMN NO. 501 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

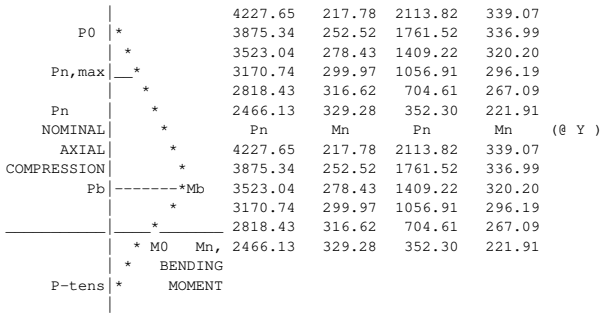
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	347.63	10.51	0.02015

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	347.63	16.09	0.03086



COLUMN NO. 502 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

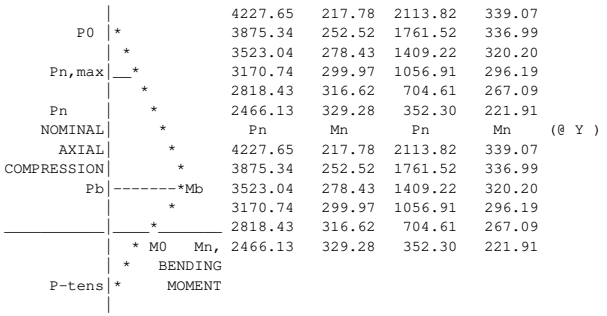
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	347.63	10.51	0.02015

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	347.63	16.09	0.03086



COLUMN NO. 583 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	222.33	4.88	0.00627

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	222.33	67.66	0.08695

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens	MOMENT				

COLUMN NO. 585 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 413.7 FC - 35.0 MPA, SQRE SIZE - 400.0 X 400.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 1600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
8 - 20 MM	1.571	101	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	495.00	10.77	0.01450

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
5724.94	4579.95	1928.86	343.44	178.1
M0	P-tens.	Des.Pn	Des.Mn	e/h
170.82	-1039.71	495.00	17.54	0.02363

	Pn	Mn	Pn	Mn	(@ Z)
	4227.65	217.78	2113.82	339.07	
P0	3875.34	252.52	1761.52	336.99	
	3523.04	278.43	1409.22	320.20	
Pn,max	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
Pn	2466.13	329.28	352.30	221.91	
NOMINAL	Pn	Mn	Pn	Mn	(@ Y)
AXIAL	4227.65	217.78	2113.82	339.07	
COMPRESSION	3875.34	252.52	1761.52	336.99	
Pb	3523.04	278.43	1409.22	320.20	
	3170.74	299.97	1056.91	296.19	
	2818.43	316.62	704.61	267.09	
M0 Mn,	2466.13	329.28	352.30	221.91	
BENDING					
P-tens	MOMENT				

**********END OF COLUMN DESIGN RESULTS**********

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224. END CONCRETE DESIGN
225. FINISH


          ********** END OF THE STAAD.Pro RUN **********

            **** DATE= JAN 29,2020   TIME= 12:28:48 ****

      ************************************************************
      *   For technical assistance on STAAD.Pro, please visit    *
      *   http://selectservices.bentley.com/en-US/               *
      *                                                          *
      *   Details about additional assistance from               *
      *   Bentley and Partners can be found at program menu      *
      *   Help->Technical Support                                *
      *                                                          *
      *      Copyright (c) 1997-2014 Bentley Systems, Inc.       *
      *                http://www.bentley.com                    *
      ************************************************************
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ATTACHMENT 5 COMMENT RESOLUTION LOG SHEET / COMPLIANCE REPORT

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Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.: 216**

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE: BECHTEL-WTRAN-001488, 2 March 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 2 March 2020	Comment by	EPC Response DATE: 06 March 2020	Status	Remarks
			General Comments (cc)				
MD-216-49WF-EG-CV-CAL-1001-A01	49WF	1	The term "Kn" to be replaced with "kN" in page no 9.	Bechtel	The same is incorporated in the design document	Closed	
MD-216-49WF-EG-CV-CAL-1001-A01	49WF	2	Define Ton, i.e. 2000 lbs or 1000 kg.	Bechtel	1 Ton is equivalent to 1000 kg.	Closed	